

RIBEIRINHO REDD+ PROJECT

MONITORING REPORT

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1 SUMMARY OF PROJECT BENEFITS

The project protects over 199,962 ha in Portel, Para, ranked as a region depending on the year, either the 1st or 2nd highest rate of deforestation in the Amazon. The project goal is to transform Portel, Para one of the poorest, lowest educated regions of Brazil into a stable, developed region that depends on sustainable forest use for its economy. The project knows that the only way to completely end illegal logging is to have the ribeirinhos stable so they can be the ones that can state “No more”. The project working with neighboring REDD projects plans to build 30 additional high quality schools in the region, that will be used for children education, adult training, and university laboratories. These 30 additional high quality schools, health clinics and security offices are above and beyond the 30 schools, health clinics and security offices that are being built by the Rio Anapu-Pacaja REDD Project.

1.1 Unique Project Benefits

Outcome or Impact	Achievements during the Monitoring Period	Section Reference	Achievements during the Project Lifetime
1.) Preservation of 199,962 hectares	The project has been able to preserve the land during the verification period. Despite a far more aggressive effort by illegal loggers. The project has provided protection and conservation to 199,962 hectares of private property land in a highly unstable municipality of Portel, Para. This has resulted in the protection of flora and fauna across a large area.	2.1.1	Same as Monitoring Period.
2) Cook stove distributed to community people	The project has distributed around 500 eco-stoves to the communities in the periphery of the project area.	2.1.1	Same as Monitoring Period
3) Land Demarcation to prevent dislocation and eviction: land tenure related issues solved	875 families as part of this project have directly gained land tenure certificates. Despite the illegal loggers stating the government certificate is not valid and being hostile to the riverine and the project.	2.1.1	Same as Monitoring Period
4) Sustainable Agro-forestry Technics	50 people benefitted in learning about alternative cash crops which will improve the economy once they use this training and implement projects. This helps the local population and transitions them	2.1.1	Same as Monitoring Period

	away from nearly solely relying on Cassava production which has very little economic benefit. The project has built up the local economy by focusing on Jatai medical grade honey production, which is a high dollar product that greatly improves the local economy without requiring deforestation. The project started in 2020 is delivering bee keeping equipment.		
5) Bio-diversity Monitoring	50 people were trained to monitor the fauna that would enter the area in the vicinity of their homes. They were guided to use their cell phone cameras to take pictures and record the animals that they saw.	2.1.1	Same as Monitoring Period

1.2 Standardized Benefit Metrics

Category	Metric	Achievements during Monitoring Period	Section Reference	Achievements during the Project Lifetime
GHG emission reductions & removals	Net estimated emission removals in the project area, measured against the without-project scenario	N/A		-
	Net estimated emission reductions in the project area, measured against the without-project scenario	4,042,310	3.2.4	33,927,167
Forest ¹ cover	For REDD ² projects: Number of hectares of reduced forest loss in the project area measured against the without-project scenario	17,565 ha	3.2	199,962 Ha

¹ Land with woody vegetation that meets an internationally accepted definition (e.g., UNFCCC, FAO or IPCC) of what constitutes a forest, which includes threshold parameters, such as minimum forest area, tree height and level of crown cover, and may include mature, secondary, degraded and wetland forests (*VCS Program Definitions*).

² Reduced emissions from deforestation and forest degradation (REDD) - Activities that reduce GHG emissions by slowing or stopping conversion of forests to non-forest land and/or reduce the degradation of forest land where forest biomass is lost (*VCS Program Definitions*).

Category	Metric	Achievements during Monitoring Period	Section Reference	Achievements during the Project Lifetime
	For ARR ³ projects: Number of hectares of forest cover increased in the project area measured against the without-project scenario	NA		-
Improved land management	Number of hectares of existing production forest land in which IFM ⁴ practices have occurred as a result of the project's activities, measured against the without-project scenario	NA		-
	Number of hectares of non-forest land in which improved land management has occurred as a result of the project's activities, measured against the without-project scenario	10,000 hectares	4.3.1	Same as Monitoring Period
Training	Total number of community members who have improved skills and/or knowledge resulting from training provided as part of project activities	50 families	4.3	Same as Monitoring Period
	Number of female community members who have improved skills and/or knowledge resulting from training provided as part of project activities	50	4.3	Same as Monitoring Period
Employment	Total number of people employed in of project activities, ⁵ expressed as number of full time employees ⁶	11	4.3	Same as Monitoring Period

³ Afforestation, reforestation and revegetation (ARR) - Activities that increase carbon stocks in woody biomass (and in some cases soils) by establishing, increasing and/or restoring vegetative cover through the planting, sowing and/or human-assisted natural regeneration of woody vegetation (*VCS Program Definitions*).

⁴ Improved forest management (IFM) - Activities that change forest management practices and increase carbon stock on forest lands managed for wood products such as saw timber, pulpwood and fuelwood (*VCS Program Definitions*).

⁵ Employed in project activities means people directly working on project activities in return for compensation (financial or otherwise), including employees, contracted workers, sub-contracted workers and community members that are paid to carry out project-related work.

⁶ Full time equivalency is calculated as the total number of hours worked (by full-time, part-time, temporary and/or seasonal staff) divided by the average number of hours worked in full-time jobs within the country, region or economic territory (adapted from UN System of National Accounts (1993) paragraphs 17.14[15.102];[17.28]).

Category	Metric	Achievements during Monitoring Period	Section Reference	Achievements during the Project Lifetime
	Number of women employed in project activities, expressed as number of full-time employees	3	4.3	Same as Monitoring Period
Livelihoods	Total number of people with improved livelihoods ⁷ or income generated as a result of project activities	600	4.3	Same as Monitoring Period
	Number of women with improved livelihoods or income generated as a result of project activities	330	4.3	Same as Monitoring Period
Health	Total number of people for whom health services were improved as a result of project activities, measured against the without-project scenario	400	4.3	Same as Monitoring Period
	Number of women for whom health services were improved as a result of project activities, measured against the without-project scenario	220	4.3	Same as Monitoring Period
Education	Total number of people for whom access to, or quality of, education was improved as a result of project activities, measured against the without-project scenario	300	4.3	Same as Monitoring Period
	Number of women and girls for whom access to, or quality of, education was improved as a result of project activities, measured against the without-project scenario	180	4.3	Same as Monitoring Period
Water	Total number of people who experienced increased water quality and/or improved access to drinking water as a result of project activities, measured against the without-project scenario	2,500	4.3	Same as Monitoring Period

⁷ Livelihoods are the capabilities, assets (including material and social resources) and activities required for a means of living (Krantz, Lasse, 2001. *The Sustainable Livelihood Approach to Poverty Reduction*. SIDA). Livelihood benefits may include benefits reported in the Employment metrics of this table.

C a t e g o r y	Metric	Achievements during Monitoring Period	Section Reference	Achievements during the Project Lifetime
	Number of women who experienced increased water quality and/or improved access to drinking water as a result of project activities, measured against the without-project scenario	1,300	4.3	Same as Monitoring Period
Well-being	Total number of community members whose well-being ⁸ was improved as a result of project activities	5,000	4.3	Same as Monitoring Period
	Number of women whose well-being was improved as a result of project activities	2,500	4.3	Same as Monitoring Period
Biodiversity conservation	Change in the number of hectares significantly better managed by the project for biodiversity conservation, ⁹ measured against the without-project scenario	10,000	2.1.1	Same as Monitoring Period
	Number of globally Critically Endangered or Endangered species ¹⁰ benefiting from reduced threats as a result of project activities, ¹¹ measured against the without-project scenario	5	5.1	Same as Monitoring Period

2 GENERAL

2.1 Project Description

2.1.1 Implementation Description

The project has been fully operational since 2017. The PD is under validation and verification process and 199,962 hectares are conserved as native forest land.

⁸ Well-being is people's experience of the quality of their lives. Well-being benefits may include benefits reported in other metrics of this table (e.g. Training, Employment, Health, Education, Water, etc.), but could also include other benefits such as empowerment of community groups, strengthened legal rights to resources, conservation of access to areas of cultural significance, etc.

⁹ Biodiversity conservation in this context means areas where specific management measures are being implemented as a part of project activities with an objective of enhancing biodiversity conservation.

¹⁰ Per IUCN's Red List of Threatened Species

¹¹ In the absence of direct population or occupancy measures, measurement of reduced threats may be used as evidence of benefit

The project has given out 500 cook stoves in between 2017 to 2020. These cook stoves have a tremendous benefit in the region for several reasons. The main reason for the cookstoves is due to a reduction in smoke and this helps the people cooking prevent respiratory issues. The other reason is that the eco-stoves use wood scraps instead of open fires.

The project goal is to demonstrate the economic feasibility of preserving the forest in regions that are already heavily degraded, and where it is still considered socially acceptable to convert forest to agriculture. Prior to the current owner, the land was a degraded forest, with no cutting plan. The forest area still has a high threat situation, and it is common for illegal loggers to be operating in and around the project area.

As discussed more thoroughly in Section 3.2, no further leakage monitoring is required. Permanence is addressed through the risk buffer. A Non-Permanence Risk Report was prepared under VCS 3.2 and provided to the Verifier. After buffer calculation, a total of 4,042,310 VCUs have been generated from the start date of the project until June 30th, 2021 (current monitoring period).

2.1.2 Project Category and Activity Type

The project corresponds to the VCS Scope VM00015 – for Unplanned Emissions from Deforestation and forest Degradation. The project aims to protect rainforest, which are expected to be deforested in the absence of the Project.

Project Scope 14: Agriculture, Forest, and other Land Use (AFOLU)

Project Category: Reduction Emission from Deforestation and Degradation (REDD)

Grouped Project: No

Type of Activity: Avoided Unplanned Deforestation (AUD)

This Project is planned to be registered under the Verified Carbon Standard (VCS) as a Reducing Emissions from Deforestation and forest Degradation (REDD) project and has been developed in compliance with the Verified Carbon Standard,11 Version 3. The Project will reduce emissions from unplanned deforestation.7

2.1.3 Project Proponent(s)

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Debora Pinheiro da Silva	Maria Sorlene Portilho Barbosa
Deiziane Cardoso Araujo	Maria Trindade da Cunha Lobato
Deiziane Pinheiro Fonseca	Maria Trindade Moraes Araujo
Delmiro Palheta dos Santos	Maria Valadares da Silva
Demison Pereira da Costa	Maria Vandeleia Pereira dos Santos
Demison Prata Ladislau	Mariade nazaré Barbosa de Almeida
Denildo Barbosa Soares	Marilda Batista Pereira
Denilson Salazar Alves	Marilda da Silva Brabo
Denis de Freitas Palheta	Marilda dos Santos Correa
Denise Barbosa Gibson	Marilene Aires da Costa
Derlei de Almeida da Silva	Marilene da Silva Guedes
Deusiane Barbosa Gibson	Marilene dos Santos Souza
Deuza Maria da Silva Lobo	Marilene Silva Pires
Deuza Rodrigues Pereira Moura	Marilenem Fonseca do Nascimento

Deuzarina Noronha Gomes	Marinaldo Evangelista da Cruz
Deuzerene Vaz Souza Rodrigues	Marinalva Leao Maia
Diego da Cruz Gomes	Marineia de Aquino Vilela
Dielciane Pinto da Costa	Marineide Valente Pimental
Diemilly Brabo Mendes	Mario Barbosa de Assis Filho
Dileia Barbosa de Novais	Mário de Nasare Goes puresa
Dilma de Almeida da Silva	Marivani Pinheiro Martins
Dimilson Primavera Almeida	Marlene da Costa Almeida Moraes
Dinair Gonçalves Santana	Marlene de Aquino Vilhena
Dinalva Lobato Alves	Marlene Duarte Pantoja
Dinalva Mendes da Silva	Marlene Mavilha dos Santos
Dinelson Baliero Valente	Marlene Santana
Dione Mendes Santana	Marli Pimentei dos Santos
Dionizia Santana Alves	Marliete Carvalho Rocha
Djalma de Freitas Braga	Marlison Jardim de aquino
Doberto Rodrigues Saraiva	Marta Barbosa Pantoja
Dolores de Sousa Correa	Matheus da Silva Nascimento
Domingas de Freitas Mavigno Rodrigues	Mauricio de Almeida Braga
Domingas Santos da Silva	Mauriete do Nascimento Serrao
Domingos dos Santos Ramos	Mauro Lopes Ferreira
Domingos Gomes dos Santos	Mauro Santos Silva
Domingos Pereira	Maycon Damasceno Resendo
Domingos Pinheiro Gama	Maycon Valente Moraes
Domingos Rodrigues da Costa	Melquizedeque de Moraes Lobo
Doralice Jardim Correa	Miguel Almeida de Lima
Doriedina Melo Palheta	Miguel Brabo da Silva
Douglas Valente Lopes	Miguel de Oliveira Fonseca
Dulcileia Pimental Barros	Miguel Grana da Silva
Dulcirene Silva Pires	Miguel Machado da Silva
Edanilo Souza da Silva	Miguel Santana Barbosa
Edemilson Vilarino dos Santos	Milena Santana Lobato
Edemir Crisostomo de Oliveira Santos	Miquel da Silva valadares
Edenilson de Freitas Braga	Miquel Valadares da Silva
Eder Almeida de Freitas	Miracy Curcino Bezerra Nery
Edicarlos Miranda Silveira	Miraldo Ferreira Moura
Ediel Jardim Gomes	Mirle Vilela dos Santos
Edielsa Barbosa Rodrigues	Moacyr Santos dos Santos
Edielson Barboza Valadares	Moises Alves do Nascimento
Edila Pinheiro Guerreiro	Moises de Aquino Marques
Edilamar Tavares de Aquino	Moises Ferreira Neres
Edilene Gomes Cardoso	Monica Barbosa Pimenta
Edilene Goncalves Alves	Mozaniel dos Santos Correa
Edileuza Braga Texeira	Nair Almeida Aquino
Edilson da Luz Almeida	Nairacy Moreira dos Santos
Edimar Palheta de Oliveira	Naldo Barbosa Rodrigues
Edimilson Machado Dutra	Naliele da Silva Souza

Edimilson Pacheco de Souza	Nataliene Dos Santos
Edina da Conceição Valadares da Silva	Natalina Pinheiro da Silva
Edina Maria Vaz Santana	Nayara Pimentel da Silva
Edina Oliveira da Silva	Nazael Noronha Valente
Edinalda Rodrigues de Almeida	Nazaré Brazão de Freitas
Edinaldo Barboza Rodrigues Junior	Nazareno do Nascimento
Edinaldo da Silva Pacheco	Nazir Sanches Mendes
Edinaldo da Silva Soares	Nelita de Jesus da Costa Almeida Moraes
Edinalva Alves Gomes	Nelsinho Pena Barbosa
Edinelson Cunha da Silva	Nelson Alves de Oliveira
Edinete Baia Machado	Nelson Gama Pantoja
Edineuza Castor Lopes	Neosiane Brazão de Freitas
Edinho da Silva Marques	Nerias Barroso da Cruz
Edinho Moraes da Silva	Neuracir Pinheiro da Silva
Edinho Moreira Mendes	Neuziane Noronha Valente
Edinilson Pacheco de Souza	Neuziane Valente Ramos
Edinilson Tavares Miranda	Nici Laura Barbosa de Almeida
Edion Carlos Mendes da Silva	Nicileia Valente de Paiva
Edison Ancantara da Silva	Nicolau Carvalho Barbosa
Edison de Aquino Guimarães	Nila Magno Oliveira
Edison Miranda da Silveira	Nilcivani da Silva Santos
Edison Noronha dos Anjos	Nildo Lope da Costa
Edivaldo de Oliveira Boguea	Nildon de Souza Pires
Edivaldo Fonseca do Nascimento - Sitio Acaizal	Nilcleia Alves da Silva
Edivaldo Santana Alves	Nilson Marques Tupinamba
Edmilson Braga da Costa	Nilson Primavera Almeida
Edmilson da Silva Santos	Niralci Pinheiro da Silva
Edmilson Miranda Silveira	Nivaldo Marinho Gomes
Edna Cardoso da Silva	Nivaldo Santana Castor
Ednaldo Gloria do Carmo	Nizomar Gomes Santana
Ednaldo Martins	Nizomara dos Santos
Edson de Jesus Cardoso de Oliveira	Noel Gulmaraes Goncalves
Eduardo Evangelista da Cruz	Norberto Braga dos Santos
Elana de Paiva Monteiro	Ocilene Chagas Damasceno Resendo
Elcicleia Caldas de Almeida Souza	Octavio Alves Valdares Filho
elcilene Cadas Almeida	Odaia da Silva Rodrigues
Elealdo da Silva dos Santos	Odiam Correia da Silva
Elen de Jesus Gomes	Odeliane da Silva Pereira Vanzeler
Eleonora dos Anjos da Silva	Oderleu Curcino Nery
Elessandra dos Santos	Odete Santos da Silva
Eliany Noronha Valente Novais	Odimar da Silva e Silva
Elias Barbosa Marinho	Odinela Valente Pimental
Elias machado Carvalho	Odma Gomes Caldas
Elicilda dos Santos Pinheiro	Odocio Gomes Chaves
Eliel Alves Baia	Odonio Pereira da Costa
Eliel Jardim Correa	Olavo Silva dos Santos

Elielson Pinto da Silva	Oliveira Santos dos Anjos
Elielson Ribeiro Miranda	Orceleia Correa da Silva
Elielton Silva da Silva	Ordenizio Barbosa Souza
Elielza da Silva Ramos	Orisvaldo dos Santos Souza
Elieny Araujo Gomes	Orisvaldo Marcal dos Santos
Elieuzo Baia da Cunha	Orlando Barbosa
Eliezer Machado Silveira	Orlando Barbosa Junior
Elinaldo Araujo Gomes	Orlando Rodrigues da Cunha
Elinalva Rodrigues de Oliveira	Ortencia Alves de Freitas
Elinete Moraes da Silva	Osana Gonçalves Barbosa Pinto
Elisabete da Silva Santana	Oscar da Luz da Silva
Elisete da Silva Valadares dos Santos	Oseas Silva da Silva
Eliseu Barbosa Marinho	Oseias Balieiro Cardoso
Elisiane Cardoso de Oliveira	Oseias da Silva Novaes
Elivaldo Baia da Cunha	Oseias de Aquino Guimarães
Elivaldo Evangelista da Cruz	Oseias dos Santos Caldas
Elivane da Silva Silveira	Oseias Ferreira Guimarães
Elivane Franca Novas	Osmarina Ferreira Lopes
Elizabete Guedes Serrao	Osvaldo Alves Valente
Elizana Gomes Medeiro Marques	Osvanildo Veiga Martins
Elizangela Brabo Farias	Othon dos Santos Pereira
Elizangela da Luz Freitas	Ozana da Silva de Souza
Elizangela Pinheiro Pimenta da Silva	Ozana dos Santos Laranjeira
Elizangela Sousa de Almeida	Ozenias Primavera dos Santos
Elizeu de Freitas Correa	Oziamar Gomes Caldas
Eliziete da Silva Pinheiro	Ozimar da Silva e Silva
Ellem de Almeida Rodrigues	Ozinethe Baia Machado
Eloy de Freitas Monteiro	Ozivaldo Gomes Novais dos Santos
Elrilene Caldas de Almeida	Paloma Aquino Guimarães
Elsino Castelo Pereira	Patricia da Silva Ramos
Elva da Silva Silveira	Patricia de Souza Vilhena
Elva da Silva Silveira	Patricia Almeida Lobo
Elza Silva Melo	Paula de Jesus Ferreira Silva
Elzilene Caldas De Almeida	Paulo Cesar Fontes Reis
Elzinita Barbosa de Carvalho	Paulo Dias dos Santos
Elzo Machado Serrao	Paulo Marques da Trindade
Emilly Mariane Ladislau da Silva	Paulo Ronaldo dos Santos Fonseca
Enisia Ribeiro dos Santos	Pedro Almeida da Silva Filho
Eralda da Silva Costa	Pedro Barbosa Barros
Eraldo Santos dos Anjos	Pedro de Oliveira e Oliveira
Erenildo Palheta de Melo	Pedro Farias Ponte
Erica da Silva Vieira	Pedro Jardim
Erica Pinto da Silva	Pedro Maia de Carvalho
Erika Pinheiro de Souza	Pedro Pereira Pinto Neto
Eristiana dos Santos Caldas Novais	Pedro Pereira Tavares
Eriton Adeudato Cardoso	Pedro Rodrigues de Oliveira

Errilaudo Moraes de Brito	Petrolina Alves de Freitas
Ertorgio Rodrigues da Cunha	Rael Lopes dos Santos
Esladeck da Costa Carvalho	Rafael Goes Pantoja
Esmeralda Valente da Silva	Rafaela Moura dos Santos Souza
Esperidião Martins da Silva	Railane Lopes de Lima
Estelita Camara da Cunha	Raimunda Alves de Oliveira
Etevaldo Pereira da Silva	Raimunda da Trindade Moraes Sales
Eulaziany Ferreira de Souza	Raimunda dos Santos Farias
Eullem de Melo Brabo	Raimunda Elivane Pinheiro Carvalho
Evandro Araujo gomes	Raimunda Rodrigues de Lima
Evandro Gonçalves de Abreu	Raimunda Valente
Evanilson da Silva Santos	Raimundo Almeida dos Santos
Fabiana Pereira dos Santos	Raimundo Angelo Alves da Silva
Fabio do Nascimento Palheta	Raimundo Baia de Oliveira
Fabio Junior Barbosa Pinto	Raimundo Correa Baia Junior
Faustina Oliveira dos Santos	Raimundo da Silva Goncalves
Felicia de Moraes Lobo Otone	Raimundo da Silva Pereira
Feliciano Correa Palheta	Raimundo Do Socorro Adeodato Cardoso
Fernando de Souza Valente	Raimundo do Socorro Ferreira da Costa
Filomena Vilhena Santana	Raimundo do Socorro Galena dos Santos
Floriano dos Santos Marques	Raimundo Farias Pantoja
França Gomes da Silva	Raimundo Goes Pantoja
Franciclei Duarte da Silva	Raimundo Jorge Monteiro de Albuquerque
Francidalva Barrozo Salazar	Raimundo Lobato da Silva
Francidalva Machado Rodrigues	Raimundo Magno Negroao
Franciete Santos dos Santos	Raimundo Mago Moraes Brabo
Francigel Duarte da Silva	Raimundo Marçal dos Santos
Francinildo Barbosa Dias	Raimundo Matos Rodrigues
Francisca Alves Jardim	Raimundo Mendes Pereira
Francisca Alves Pimentel	Raimundo Nonato
Francisca de Oliveira Lobato	Raimundo Nonato da Silva
Francisco Costa da Silva	Raimundo Nonato Marques Tupinamba
Francisco da Cruz Barros	Raimundo Oliveira Fonseca
Francisco de Almeida Rodrigues	Raimundo Pacheco de Azevedo
Francisco de Freitas Cunha	Raimundo Pantoja Baia
Francisco do Nascimento Moraes	Raimundo Pereira de Melo
Francisco dos Santos Marques	Raimundo Pereira Valente
Francisco Jardim Pereira	Raimundo Pinheiro Ladislau
Francisco Mauro de Almeida	Raimundo Rodrigues
Francisco Pereira Viana	Raimundo Rodrigues dos Santos
Francisco Rodrigues Lopes	Raimundo Trindade do Nascimento Moraes
Francisco Wilagaissom Lemos Rabelo	Raquel da Silva Ferreira Costa
Franklin Silva de Souza	Raquel de Melo Farias
Frederico da Silva Fereira	Raquel Pacheco de Lima
Gelson Braga Alves	Rasangela Maia Reis Macedo

Genilson Souza Brabo	Raysa da Costa Castor
Geovane Barbosa de Novais	Rebeca Aline Ferreira Rebelo
Geovane Brazão de Freitas	Reginaldo Carvalho Neves
Geovane da Gama Lopes	Reinaldo Evangelista da Cruz
Gercika Rocha da Gama	Renato Rodrigues Saraiva
Gerciolina Santana da Cruz	Renato Sanches Gomes
Gerison Duarte Nascimento	Ricardo Rodrigues da Cunha
Gesival Fernandes Reis	Rildo da Silva e Silva
Gideoo Santana Barbosa	Risoneia da Cruz Valadares
Gilberto da Silva Valente	Rita alves do Nascimento
Gilmar da Silva Valente	Rizonete Costa Pinheiro
Gilvandro Jose Rodrigues	Robert dos Santos de Souza
Giovan Maia Serrao	Roberto Bento Monteiro de Albuquerque
Gisele Brabo Farias	Roberto da Silva Carvalho
Gracilene Primavera Almeida	Robson de Souza Rodrigues
Gracimar Fonseca Mavigno da Cruz	Rodrigo Aquino Guimarães
Guilherme Dias Dos Santos	Rodrigo Palheta Moura
Guilherme Fonseca Ferreira	Rodrigo Valente da Silva
Hailton Diniz Serrão	Rogério Prata de Lima
Hamilton de Freitas Braga	Romario da Silva Castor
Hamilton Ferreira Moura	Romario Nobre Lobato
Hamilton Lima Malaquias	Romario Rodrigues da Silva
Harleson Valente de Aquino	Rômulo de Aquino Martins
Helca Monteiro Cunha	Ronaldo Barbosa Rodrigues
Heleilson Machado de Aquino	Roney Jardim Moraes
Heleno Valente Rodrigues	Ronis Tupinamba Barros
Helio Jardim Martins	Ronivaldo de Araujo dos Santos
Henrique Alves Martins	Rosa Benicia da Costa
Henriqueta Jardim Martins	Rosa da Silva Cunha
Herberson Ribeiro da Silva	Rosa Lobato
Hermitonn Palheta de Oliveira	Rosa Maria Cordovil de Melo_Sítio Limoeiro
Honorato Santos da Silva	Rosalino Ferreira dos Santos
Humberto dos Santos Duarte	Rosana Gomes Costa
Ilson Primavera Almeida	Rosana Gomes Farias
Ipiranga Juniel Noronha Gomes	Rosana Machado Maia
Iraci Brabo Fernandes	Rosangela Fereira Lopes
Iracilda Dias Dos Santos	Roseane de Almeida Sousa
Iracilda Ferreira Machado	Roseane Marcolino Moraes
Iracilda Pacheco	Roseane Valentin da Silva
Iraelson da Silva Soares	Roseli de Freitas Palheta
Iranildo da Silva Santos	Roselin de Azevedo dos Santos
Iranildo Mercês da Silva	Roselma da Costa do Carmo
Irenildo Gomes Pimental	Rosenice Rodrgiues de Araujo
Iriane da Silva Soares	Rosenilda Sanches Baia
Irineu Mesquita Lobato	Rosenildes Lobato Correa da Silva

Isis de Fatima Araujo dos Santos	Rosenildo da Luz Almeida
Ismael da Silva Rodrigues	Rosiane Ramos Gomes
Ivanete de Almeida da Silva	Rosiangela Conceição Santos
Ivanete de Freitas da Silva	Rosilene Dias dos Santos
Ivanildo da Silva Rodrigues	Rosilene dos Santos Souza
Ivanilson Machado Silveira	Rosilene Gomes Alves
Ivanilza do Nascimento Mathias	Rosilene Machado Silveira
Izabel Costa Pinheiro	Rosinaldo Barbosa Rodrigues
Izaías Primavera dos Santos	Rosineia Rodrigues Toscano
Izanildo da Costa Pinheiro	Rosineia Silva de Freitas
Izaurenis Goncalves Carvalho	Rosineide Nunes Ferreira
Izomar Barbosa de Almeida	Rosinete Barbosa De Carvalho
Izonete Gomes de Souza	Rosinete Gouveia Farias
Jacileia da Costa Carvalho	Rosinete Leite Bandeira
Jacileia do Nascimento Serrao	Rosinete Pacheco dos Santos
Jaciro Lobato da Silva	Rosires Monteiro Loureiro
Jacqueline Barbosa Souza	Rosivaldo Araujo do Carmo
Jacquele Barbosa Souza	Rosivaldo Barbosa Barros
Jaidemilson Gomes de Souza	Rosivaldo Martins
Jailene Souza da Silva	Roziane Rodrigues Moura
Jainha de Souza Baia	Rozilda Oliveira Moura
Jair Barbosa Maia	Rozinaldo Araujo dos Santos
Jair Miranda da Silva	Rubemax Brabo Resendo
Jairo Barboza Alves	Rubiane Gomes da Cruz
Jairo da Costa Brito	Rui Santana
Jairo da Cruz Valadares	Ruthi da Silva Marcolino
Jairo Gomes de Souza	Sabina Pacheco da Silva
Jairo Serrao Rodrigues	Sabino Barbosa Rodrigues
Jairzinho Duarte dos Santos	Sabrina Rocha Monteiro
Janari dos Santos Prata	Salomao Ferreira Mathias
Jaqueline Lorena Cunha Pontes	Samara Raposo de Oliveira
Jaqueline Miranda Alves	Samara Silva e Silva
Jardeane Tenorio Barbosa	Samarina Pontes Maia
Jardel Braga Barbosa	Samuel dos Santos Filho
Jardel Brazao de Freitas	Samuel Pontes Maia
Jardel de Aquino Martins	Sandra do Socorro Silva Gama
Jardel Fonseca Jardim	Sandra Helena Maciel Freitas
Jaurimar Alves de Vilela	Sandra Maria de Matos da Costa
Jean Carlos Souza Alves	Sandra Muniz Valente
Jeferson Almeida da Cruz	Sandro Rodrigues Gomes
Jeferson Souza Alves	Sansão Primavera dos Santos
Jeila Souza Alves	Sebastiana Almeida Palheta
Jeny Lobo Moraes	Sebastiana Barbosa de Souza
Jeosirene Pacheco da Silva Santos	Sebastiana dos Santos Souza
Jeremias de Moraes Lobo	Sebastiao Lopes de Oliveira Filho
Jerfeson Rodrigues Gomes	Sebastiao Magno Marques

Jessica Azevedo Galvão	Sebastiao Nascimento da Silva
Jesus Araujo Gomes	Sesbastião Martins Alves
Jhemission Santos Monteiro	Sidelma Ferreira Pacheco
Jhon Carlos de Jesus Almeida	Sidnei Primavera Braga
Jhone Aquino Guimarães	Siely Almeida Ramos
Joa Henrique Goncalves Lobato	Silelda Santiago de Lima
Joabe da Silva Conceicao	Silene Mendes da Silva
Joana Alves de Souza	Silvana da Silva Pontes
Joana de Aquino	Silvana de Aquino Guimarães
Joao Almeida Palheta	Silvany de Souza Rodrigues
João Alves Bandeira	Simar Monteiro
João Bastista Araujo Melo	Simone Barbosa de Souza
Joao Castelo de Carvalho	Simone da Silva da Costa
João de Aquino Neto	Simone dos Santos Moraes
João de Nazaré Serrão Fonseca	Simone Fernandes Guimaraes
Joao do Nascimento Moraes	Simone Jardim de Oliveira
João do Socorro Alves de Souza	Simone Machado Soares
João Ferreira de Sousa Sobrinho	Simone Serrao Guedes
João Ferreira dos Santos	Sinair Vilarinho da Cruz
João Gomes	Sinei Viana
Joao Mendes da Silva	Socorro Guilherme Palheta
João Santiago de Lima	Solange Moura Ribeiro
Joao Soares da Costa	Solomao de Sousa Silva
Joaquim Moreira da Silva	Sonia Correa da Silva
Joaquim Moreira da Silva Filho	Sonia Maria Balieiro Valente
Joaquim Vasconcelos Ferreira	Sonia Pimental Raposo
Joel Ferreira Pacheco	Sorajane Souza Alves
Joelson Ferreira Pacheco	Suelem de Oliveira Duarte
Joelson Vilhena do Nascimento	Taianny Gonçalves Santana
Jonas Mendes da Silva	Taina Gonçalves Santana
Jonas Mendes Santana	Tais Pereira Coelho
Jonas Santana Barbosa	Tatiane Monteiro Batista
Jonatas Farias Pontes	Tatianny Gonçalves Santana
Jones Miranda da Silva	Tatima Dias Vito
Jorge dos Santos Vilela	Teresa Araujo dos Santos
Jorge Matos Alves	Tereza Barbosa dos Santos
Jorge Santana Castor	Tereza de Alcantara Valente
Jose Adriano Salazar da Gama	Terezinha Barbosa Pimental
Jose Alaerco da Silva Guedes	Terezinha Guedes da Silva
Jose Albedi	Thiago de Aquino Guimarães
Jose Assuncao Araujo	Tiago Barbosa Pinto
José Benedito Noronha Gomes	Tiago Bispo Costa
Jose de Jesus Portilho dos Santos	Tiago Lima de Almeida
Jose de Moraes Sales	Tito Cordeiro da Silva
Jose de Souza Moraes	Valda de Souza Pimenta
Jose Ederaldo Costa Barbosa	Valdeci de Almeida da Silva

Jose Elizonilson Machado	Valdecir Gama de Aquino
Jose Jeovane de Almeida Barbosa	Valdeir Felix Araujo
Jose Junior de Freitas Monteiro	Valdelir Cordeiro da Costa
Jose Maria Alves de Oliveira	Valdemir Cordeiro da Costa
José Maria Barbosa Gibson	Valdemir Gibson Coelho
Jose Maria Bezerra Duarte	Valdemir Jardim Moraes
Jose Maria Candido de Oliveira	Valdenir da Costa e Silva
Jose Maria Cardoso Novais	Valdenir Valente de Paiva
José Maria Ferreira de Aquino	Valdete Gomes Carvalho
José Maria Lopes Freitas	Valdileia de Souza Gama Farias
Jose Maria Pereira Moraes	Valdineia Pureza Valente Pimentel
Jose Miguel Barbosa da Silva	Valdir Barbosa da Silva
Jose Miguel dos Santos Araujo	Valdir Cordeiro da Costa
Jose Milton Serrao Guedes	Valmir Marinho Valadares
Jose Moraes Araujo	Vanael Araújo dos Santos
Jose Moraes Serrao	Vander Correa Santiago
Jose Moreira da Costa	Vanderlei Moura dos Santos
José Orlando Mavigno de Souza	Vaneide Pereira Farias
Jose Raimundo de Souza Baia	Vaneize Noronha Gomes
Jose Raimundo dos Santos Araujo	Vanilda da Silva Silveira
Jose Raimundo Ferreira Batista	Vanilson Primavera Almeida
Jose Ribeiro da Costa	Verdilho Valadares da Silva
Jose Roberto Lopes Freitas	Victor Miranda dos Santos
Jose Santana Castor	Vilma Maria Moura dos Santos
Jose Wilke Coelho Costa	Vital Cordeiro e Silva
Joseane Rodrigues Costa	Vitória Machado de Aquino
Joselino Alves Braga	Volnaldo Barros Nascimento
Josely Machado de Aquino	Waldelei Santa Baia
Josenilton Silva Moreira	Waldelina Ferreira
Josias Alves do Nascimento	Waldir Baia Barbosa
Josias Costa Araujo	Waldir Gibson Coelho
Josias Ferreira da Silva Oliveira	Waldir Lucas Cardoso
Josiel da Carvalho Rocha	Walter Barros Nascimento
Josimar da Silva Moreiras	Walter Ferreira Alves
Josimar Duarte Fonseca	Walter Nunes de Oliveira
Josimar Pacheco Baia	Walter Willer Santos Nascimento
Josinaldo Barbosa Rodrigues	Wandeleia Santana Castor
Josinaldo Pacheco Baia	Wellington Luiz de Oliveira Chaves
Josinaldo Santos da Silva	Willibran Raatz Barcellos
Josuel de Carvalho Rocha	Zacarias Alves dos santos
Jozete Alves da Costa	Zamor Moreira dos Santos
Jucelino de Freitas Silva	Zaniela Sardes Correa
Jucelino Junior Souza Alves	Zeni Marinho da Silva
Jucelino Ramos da Silva Filho	Zenilda Barbosa Farias
Juciel Almeida Lobo	Zuleide Barbosa Souza

2.1.4 Other Entities Involved in the Project

Title	City Council Members and the Secretary of Education
Address	Avenida Duque de Caxias number 813, Bairro: Centro, CEP: 66480-000

2.1.5 Project Start Date (G1.9)

The project started on April 1, 2017

Although, forest protection initiatives and activities were developed prior, setting a 2017 project start date was a conservative approach to make sure the project had enough monitoring and enforcement experience to start producing conservation results.

No actual help to the riverine people and traditional rural villagers occurred prior to 2017 in these areas.

2.1.6 Project Crediting Period (G1.9)

30 years starting from 1st April 2017 and ending to 31th March 2047. Baseline will be renewed for every 10 years after the project start date. First baseline period renewal will be from 1st April 2027.

2.1.7 Project Location

The project is in northwest of Brazil, in the State of Para, micro region of Portel, municipality of Portel (Figure 1). The regular main transport to arrive in Portel is by boat, the trip takes approximately, 8 hours from Belem. Some parts of the area are also accessible by car, as the network of roads of the Transamazonica highway (BR-230) has already connected to logging (pioneer) roads that keep developing in the project area.

To arrive at the project area people must take a ferry from Belem to Portel, which takes from 8 to 14 hours. There are two options for ferry, one in the evening at 6 PM arriving the next morning. The other option is a day ferry at 8 AM arriving 8 hours later. From the city of Portel the project boat will take people out to the land. The closest point is 5 hours away in the project boat, or in a speed boat about 2.5 hours away.

The land of the Project (LP) has 205,820 ha and the Project Area (PA) has 199,962ha (KML file has been submitted separately).

In total, the project area is constituted by 36 different individually titled properties registered in the registry office of Portel, Para.

All properties have certified Maps with the Federal Land agency known as INCRA.

The Leakage Management Area is constituted as the Riverine communities and traditional rural villagers that are without title but are neighboring to the project area.

Figure 1: Project area – 199,962hectares

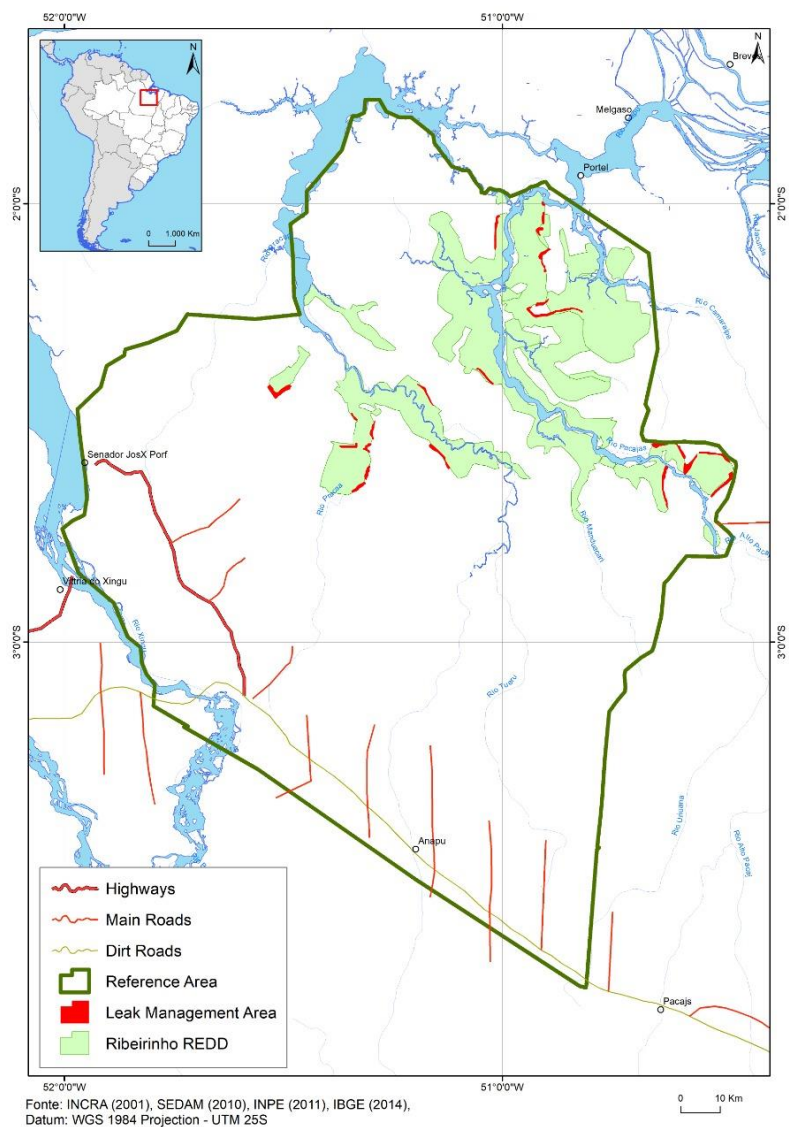


Figure 2: Satellite Picture of Project Area

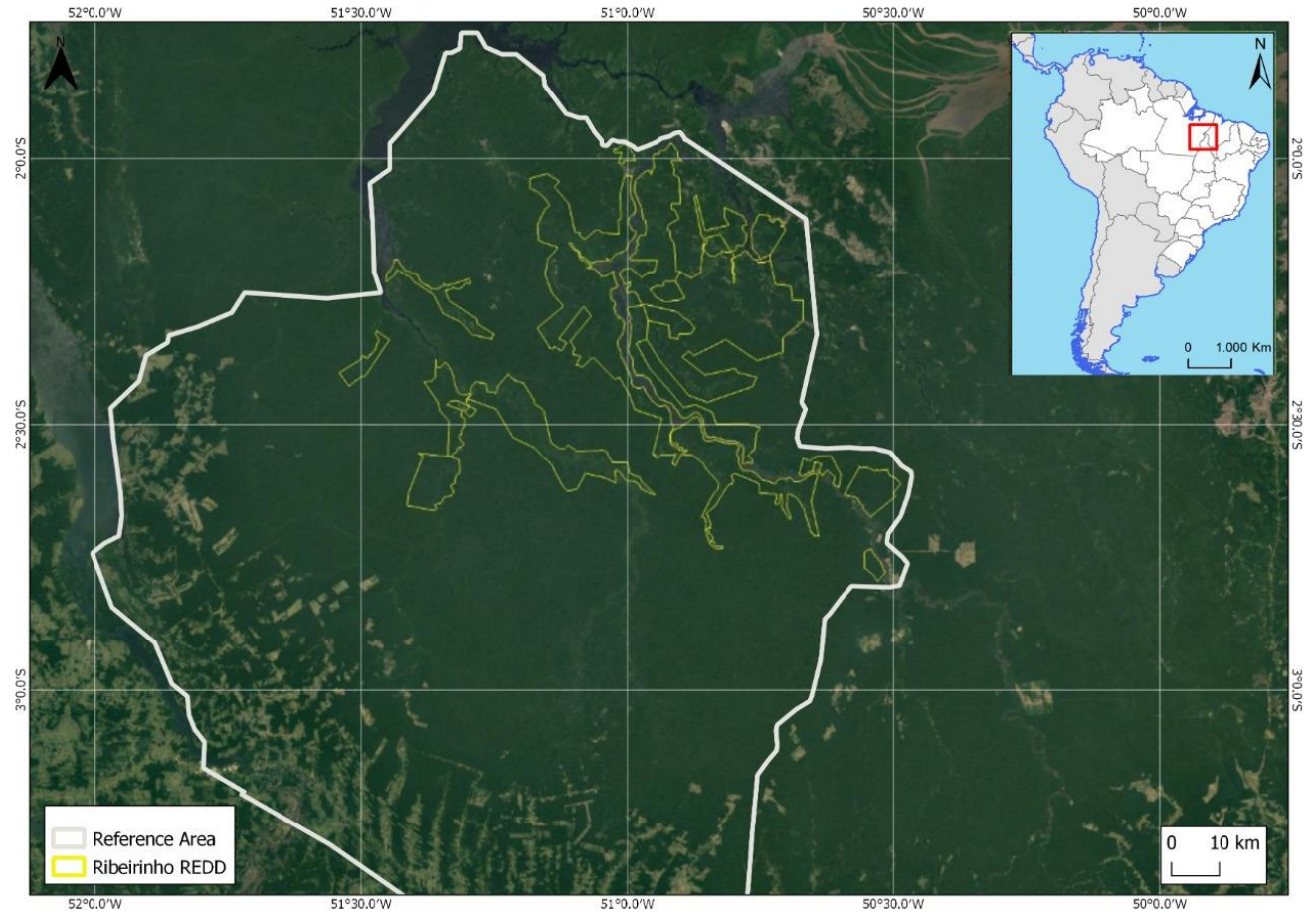
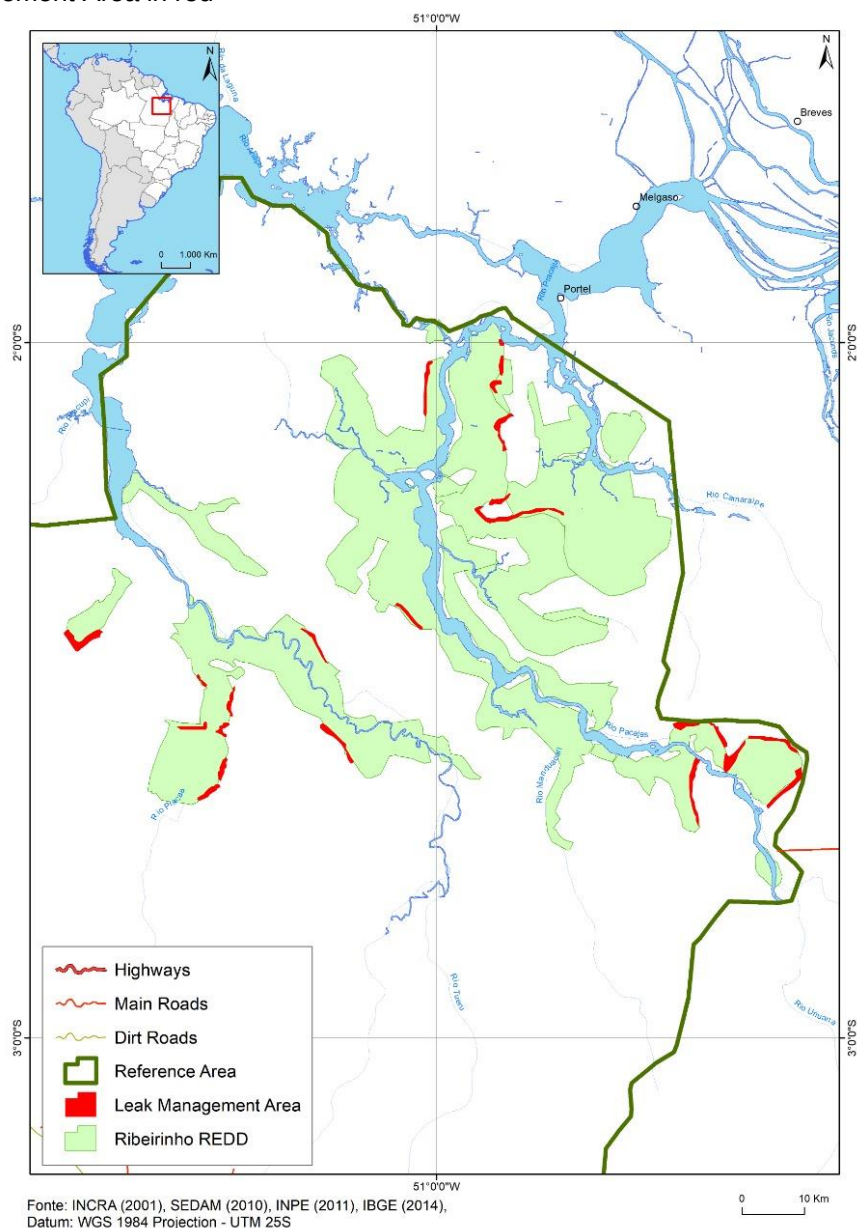


Figure 3: Leakage Management Area – 16,503 hectares

Project Area in Green

Leakage Management Area in red



2.1.8 Title and Reference of Methodology

The project used the VCS-approved methodology VM0015, v1.1, “Methodology for Avoided Unplanned Deforestation” for quantification of GHG emission reductions and removals generated from avoided unplanned mosaic deforestation. In combination with the methodology, the latest version of the following approved tools and modules were used by the project:

(Ref: <https://verra.org/wp-content/uploads/2018/03/VM0015-Avoided-Unplanned-Deforestation-v1.0.pdf>).

Tools and Modules applied:

- VCS Tool VT001 Version 3.0 – Tool for demonstration and Assessment of Additionality in AFOLU Project Activities.
- Tool for AFOLU non-permanence risk analysis and buffer determination," Version 3.2

2.1.9 Other Programs (G5.9)

- Emission Trading Programs and Other Binding Limits: there are no other emissions trading.
- Other Forms of Environmental Credit: there are no other forms of environmental credits under which the Project is planning to be eligible to participate.
- Participation under Other GHG Programs: the Project is not considering to be validated under any other GHG program.
- Currently, there is no national or jurisdictional REDD+ program. Therefore, the project is not located within a jurisdiction covered by a REDD+ jurisdictional program in Brazil and not required to follow the VCS jurisdictional REDD+ requirements.

This section is not applicable because Brazil does not have an emissions trading program and REDD+ projects from Brazil are currently not eligible for any compliance market. In contrast, the current Project is being validated and verified to the VCS and CCBS and shall be regularly verified to both the VCS and CCBS. The issuance of Verified Carbon Units (VCUs) under the VCS Registry helps to ensure the avoidance of GHG emissions being double counted. In addition, Project Participant's finances and project portfolio are independently audited each year by a certified public accountant (CPA). The current Project is not participating in any other GHG programs other than the VCS and CCBS. The current Project is not generating any other forms of environmental credits such as biodiversity offsets, watershed protection payments, or renewable energy certificates (RECs). Furthermore, the current Project has not been rejected by any other GHG program.

2.1.10 Sustainable Development

Following the Warsaw Framework for REDD-plus adopted at the UN Climate Change Conference COP19, Brazil is the first country to voluntarily submit and have a forest reference emission level technical assessed, and also the first country to submit REDD-plus results in a technical annex to the Biennial Update Report (BUR) for technical analysis. A key component of Brazil's National Climate Change Policy is the voluntary reduction in greenhouse gas emissions. The Project is in compliance with this voluntary target because the Project is a Reducing Emissions from Deforestation and Degradation (REDD+) project. Furthermore, this compliance is demonstrated via periodic verifications of the Project.

The following are the 17 SDGs and how the Project is working towards helping to meet these voluntary SDGs:

Sustainable Development Goals	Application in Project
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 1 NO POVERTY	The project plans to over 30 years change Portel from what is a incredibly poor and impoverished region of Brazil into a stable, well-educated region. The region itself has what is known as the “Oil Curse” which is an abundance of natural resources, but all of that wealth held in the hands of the few. To reach this goal education, health and security are the three pillars of the project. It is believed that educated the children and adults in higher quality schools with working restrooms, running water and actual kitchens will help transform Portel over the next 15 years from a place that was incredibly impoverished into a flourishing, sustainable community. It will not be a fast process and there will be many hard detours along the way. However it is possible to complete. The environmental payment that will be paid monthly to all the project proponents will end a great deal of poverty in the project area. The project has already announced on the radio in Portel to the project proponents that the first environmental payment is being organized and will be paid prior to December.
 2 ZERO HUNGER	The project helps in the fight against hunger through the implementation of sustainable food production systems and resilient agricultural practices through strengthening low carbon family agriculture in the area, increasing productivity in family production units, recovering areas degraded previously by implantation of Agroforestry Systems, diversifying the agricultural production with the implantation of nurseries to supply seedlings of varied species, and generating guarantee of food security for the families.
 3 GOOD HEALTH AND WELL-BEING	The project has a nurse that works in the field and attends to the needs of the people. The project as listed will build 30 schools. In each school will be a 3-room health clinic for the local population. This health clinic is incredibly important as it will also provide virtual services. It will have medicine stocked and paid for by the project. It will be able to handle most common issues. The health clinics are already under construction and as they are completed will greatly assist in the needs of the population.
 4 QUALITY EDUCATION	Initially the project provides access and encourages education through technical and professional courses enabling better employment conditions and income, especially for women and youth. In addition, activities related to education and incentives to sustainable management practices of forest resources, reduce the spread of illegal activities, promote the appreciation of cultural diversity and contribute to the sustainable development culture. In order to achieve these objectives, the project is focused on intensifying technical assistance and rural extension services, as well as offering training aimed at production bias, social organization, corporativism, leadership and financial management, developing technical and professional skills. By mid-2023 the project will have rebuilt the entire educational system of Portel. In the region, including the Ribeirinho REDD Project and the Rio Anapu-Pacaja REDD Project there are 60 schools planned to be built in total.

5 GENDER EQUALITY 	All project activities are open and stimulated for the participation of all the residents of the acting communities, especially women, youth and marginalized people. The project is a Womans Directed REDD project and is proud to be managed by Woman for the betterment of the entire community.
6 CLEAN WATER AND SANITATION 	The project is focused on drilling 1,300 water wells. After working with neighboring REDD projects to deliver 625 water filters, it was determined that water wells are more expensive, however with the funding from increased credit pricing it made more sense for long term clean water improvements. The project is actively drilling water wells and drilling as fast as the supplier of the hand pumps can deliver pumps.
7 AFFORDABLE AND CLEAN ENERGY 	The project is conducting a pilot program to provide solar panel to 20 houses in the project area. The goal is to see the feasibility and to gain governmental support to fund this extremely expensive endeavor. The cost to provide full solar with batteries to all households in the project area is beyond the budget of the project even with increased credit prices. Ten houses have had solar panels installed on them.
8 DECENT WORK AND ECONOMIC GROWTH 	This REDD project is a major employer in Portel, for building the 30 schools/health clinics and security office, there are over 120 direct jobs for the region and another 200 indirect jobs that the project has created. For the direct employees in the social services team for the project the salaries are 30% to 50% of above the average pay for that profession in Brazil. For the local population the Acai program and the high price of the Acai Fruit is helping the local population improve and have their own personal economic growth.
9 INDUSTRY, INNOVATION AND INFRASTRUCTURE 	The project has participated in setting up one of the only non-wood related industries in Portel, Para and that has already completed the contruction of an ecological brick factory. The goal is to produce bricks for building the schools, health clinics and security offices for the REDD projects in the region. However the brick factory is already running 3 shifts a day and around the clock, with a demand for bricks locally in the city.

10 REDUCED INEQUALITIES 	The primary goal is to reduce income inequality, which is incredibly hard to do in a place where the average salary is below minimum wage and that 20% of the rural population live on less than 3 dollars per month. To do this the goal is to promote at least minimum wage for everyone and this is for the project to employ and support local businesses. By employing more people more of the local population will be making at least the minimum wage. By stopping the flagrant illegal logging, the project will ensure that these million reais salaries the illegal loggers make on the back of the local population will be greatly reduced. The project is the 2nd largest investment into the community after the government, and the government budget of 185 million reais mostly can't be explained on where most of the money is going as it is not going into the reduction of inequality.
11 SUSTAINABLE CITIES AND COMMUNITIES 	The project focuses on a sustainable self-sufficient community. The project hopes to be an example for all of Portel. Currently Portel imports nearly everything it needs; it makes nearly nothing. This importation results in all usable money to be lost to businesses in other cities. It also result in increased carbon emissions as every product needs to be barged in from Belem. The project has a goal as it and the neighboring projects build these 60 schools, health clinics and security offices to build the materials locally, in a sustainable way, that can be an example to the city so they keep the wealth of Portel in Portel and this will make Portel as a whole more sustainable The school for example are using ecological bricks made in Portel, and due to this work, already a local entrepreneur saw this example of locally made sustainable bricks and is now working to open his own brick factory for Portel.
12 RESPONSIBLE CONSUMPTION AND PRODUCTION 	Through actions that encourage the responsible exploitation of natural resources, low carbon agriculture and the recovery of degraded areas, the project promotes the conservation of natural resources coupled with socioeconomic development. Some of the main components of the Project are related to the promotion of scientific research focused on the efficient use of natural resources, seeking greater integration among the parties involved in the project and focusing on sustainable business chains, generating income and well-being for local communities and making the use of natural resources available more responsible and conscious.
13 CLIMATE ACTION 	All activities undertaken by the project aim to take action to combat climate change and its impacts through the reduction of deforestation in the project area and consequently reducing the emission of greenhouse gases, contributing directly to the Brazilian goal of reducing emissions, the project has the potential to reduce 40,349,493 tCO₂ of GHG emissions in 30 years.

 14 LIFE BELOW WATER	The Amazon has a major issue below water, and that is that the rivers of the Amazon are being over fished and depleted of species that have been present for thousands of years. One species in particular that is nearly extinct in Portel is the Manatee. This animal has two small groups remaining and the riverine population has been hunting these endangered species for food. The project has hired a Veterinarian biologist who loves nature and has started a program to help get the word out for protecting the Manatee and not to hunt it. The goal is to bring back the Manatee to all parts of Portel and not only a few isolated locations. The other focus is on the turtle and a program to breed turtles to re-introduce them into the wild. The biologist has designed a program to help the Manatee to flourish in Portel.
 15 LIFE ON LAND	The project works to use education to educate the over 6,000 people involved in this project to protect the land, protect the animals on the land and to not hunt species that are endangered or primates. The terrestrial life that will be the focus will be 1. young anteater, 2. jaguar, 3. black jaguar, 4. red jaguar, 5. ocelot, 6. peccary, 7. tapir, 8. flag anteater, 9. capuchin monkey, 10. howler monkey and 11. paca As the project gains credit sales an animal sanctuary will be set up in partnership with UFRA to help species protect, and study endangered species.

16 PEACE, JUSTICE AND STRONG INSTITUTIONS 	The riverine population have little peace or justice and the institutions in Portel are controlled by the illegal logging mafia. The project has already started a lawsuit on behalf of 70 riverine families that had their land invaded by a large industrial land grabber that wants to deforest 8,750 hectares. This only happened on July 2021. The families each had their plots in the government system of SEMA. This audacious “land grabber” found what is termed as a floating title and put it right on-top of their land to claim the area, claiming 17,500 hectares with the intent to deforest 8,750 hectares. The reason it is known of the intent to deforest is because the company doing the invading stated they are an Industrial Acai producer and because in the documents they claim the area is 50% already deforested – a trick land grabbers use in order to state: “it was already deforested” in the SEMA Governmental System. This helps them avoid fines and only gain a minor fine. The attorneys are working on both the lawsuit and working with the prosecutors office in the capital to go after these local land grabbers – criminally as the local institutions are not advocates for the population. Any criminal act against the riverine population that would simply go unpunished will be targeted by the projects attorneys in Belem and Sao Paulo and the proper denunciations will be made to the authorities.
17 PARTNERSHIPS FOR THE GOALS 	The project has made two critical partnerships in Portel, that is with the Sindicato (Union) of Rural Producers who is the legal representative for the riverine population. The 2nd partnership is with 4 of the local city council members. Also with the Municipality Secretary of Education. The project is negotiating partnerships with the ITERPA, SEMA and other state and federal agencies. The goal over time to partner with the mayors office, however the current mayor’s vice mayor is the wife of a major illegal logger in the region. Thus the project stands for principles that they are against. The project believes the next person to be mayor of Portel will be in favor of stopping illegal loggers.

2.2 Project Implementation Status

2.2.1 Implementation Schedule (G1.9)

Date	Milestone(s) in the project’s development and implementation
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2017	The community groups were contacted, and one-on-one meetings took place to gain support for the project signature goal of inserting into the government database the necessary documentation to allow each family to gain land tenure documents.
January 15 th , 2017	Initial Participatory Rural Appraisal (PRA)
March 15 nd , 2017	Stakeholder's meeting on carbon credits
April 1 st , 2017	Start date of GHG accounting Period (First Verification).
January 1 st , 2018 to July 30 th , 2018	Eco-Stoves were delivered for 20 families Some land survey work was completed for each family.
September 2 nd , 2017	Implementation of biodiversity monitoring plan
January 1 st , 2018 to July 30 th , 2018	Eco-Stoves were delivered to 20 families in the project area/
December, 2018	500 land tenure certificates from the previous 18 months of field data collection were inserted into the government database of SEMA
January 1 st , 2019 to May 30 th , 2019	Eco-Stoves were delivered to 400 families.
May 10 th , 2019	Resource Management Plan completed and signed
January 1 st , 2020 to April 20 th , 2020	Additional land survey work for each family parcel is being completed so that the data can be submitted to the government for the final step in a long process to gain land title deeds for each Riverine and traditional rural villagers' family.
May 20 th , 2021	Completion of PD and MR
June 2021	Warehouse in Portel was purchased
June 15 th , 2021	Tour the site locations with Carlos and the Secretary of Education of Portel, to do site assessment prior to school construction start.
July 2021	A brick machine to make ecological bricks arrive in Portel
July 15 th , 2021	Start school construction in location.
Year 2020 to 2045	• Development and monitoring of environmental and social management activities • Monitoring of deforestation and emissions • Monitoring of biodiversity (Fauna and Flora) and High Conservation Value Areas • Development of scientific research • Verification of credits (Selection and contracting of verification body; Production of follow-up bulletins for Verification Project; Monitoring of field audit; Registration of credits) Conducting of credit marketing processes
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2.2.2 Methodology Deviations

There are no methodology deviations.

2.2.3 Minor Changes to Project Description (Rules 3.5.6)

Not applicable.

2.2.4 Project Description Deviations (Rules 3.5.7 – 3.5.10)

Not applicable – No deviations.

2.2.5 Grouped Projects

Not applicable.

2.2.6 Risks to the Project (G1.10)

Human induced risks:

The greatest human induced risk to the Project's benefits is continued deforestation and government programs that support the invasion and subsequent deforestation of private properties, even if they are already owned by other people.

The Project Zone is undergoing large pressures for new agricultural land from both the expanding population and on-going immigration into the area from the south, west and east where poor families have already deforested for there to be thousands of small cattle farms.

The illegal loggers are sophisticated and well-funded and have their own NGO's designed to appear as legitimate NGO's that call out foul of human rights to anyone that tries to stop illegal logging and help the ribeirinhos. When police from other regions show up in Portel for raids, the illegal loggers assign people to follow and track every movement, and see exactly who the police speak to, to know who is helping them locally. They will have to try and find the cleanest names from their mafia group to make denunciations to undermine or even have the project canceled. However, the fact that the illegal loggers are attacking the projects, proves that the project is very effective at stopping the illegal loggers.

Mitigation for this risk is through the Project Activities, mainly in the form of increased protection of the Project Area, creation of new income generating activities and through the promotion of improved agricultural methods as described in the Project Description Section.

The project is embarking on massive legal spending to halt the invasion of land in the project area that is being "virtually invaded" This is when a person uses a government program, registers a survey map in the government program and proceeds to try to gain title.

Land Grabbing takes place with similar steps used by the project to help the Riverine community groups gain land titles. However, in the case of land grabbers they pay multiple straw men money for them to use the persons ID number and they round up 40 or 50 people to land grab 40 or 50 plots, each plot about 1,000 hectares. The program until 2018 allowed up to 1,500 hectares to be claimed.

The Project during this monitoring period has continued to reduce the risk of leakage, illegal logging and fire by building strong partnerships with villagers in the Project Boundaries and its vicinity thus preventing deforestation activities to start.

The project is also able to stop land claims by registering the project area land title with INCRA and SEMA government agencies, showing that it is already “taken”. Only the most brazen and corrupt will proceed to land grab that is already “taken”. Unfortunately, these brazen groups form mafias and try to undermine legitimate land title in order to create their own.

The project can halt the land grabbing by employing attorneys to aggressively target when these virtual invasions take place.

Illegal logging risks were mitigated through several measures including demarcating boundaries and posting signage, and regular patrolling.

Natural Risks

The region in which the Project is located is not generally susceptible to severe or destructive natural events. Due to the fact it is a true rainforest and falls into the inter-tropical region, thus gains heavy rains year-round. Other parts of the Amazon away from the equator have pro-longed dry seasons which causes there to be higher risk to fire. The biggest issue is actually flooding and if an area floods for more than 6 months then there can be forest die-off.

The primary types of natural events that could occur would be geologic events, pests or disease, flooding or a fire in the south side of the project area could happen if it was man-made. The area is not prone to any geologic activity and poses little to no risk to the Project. As the Project Area is a native and biodiverse ecosystem the risk from pests or disease that result in significant emissions reversal is low. The primary mitigation for this risk is to maintain the forest and ensure through monitoring that the trees and ecosystem remain healthy and intact. There can be minor seasonal flooding from the annual heavy rain season. However, the species of this area are all adapted to the hydrological cycles and are not liable to flooding. The Project Area is not mountainous, therefore, there is no risk of erosion or landslides.

The risk of human induced fire has the most potential to cause damage to the forests of the Project Area. However, the risk of significant emissions reversal is low. There have been no catastrophic fires in forests of this type in this region. Therefore, natural events have low risks to the Project’s benefits.

The VCS non-permanence risk report submitted identified such risks as minimal, which is applicable for this monitoring period.

1. Communities lack of effectiveness to control the Conservation Forest area:

The Project did provide permanent land use rights against results for conservation to those families living within the Project Boundary. Families were trained to monitor the area and to protect the forest.

With the completed census, and knowing who individually owns what, if a family breaks the Conservation Forest Area plan, it is easy to identify who they are and future benefits from the project will be curtailed if the behavior continues. The project has no way to revoke title that has been provided the riverine people and traditional rural villagers. Around 875 (Cadastro Ambiental Rural) CARs were registered for this monitoring period.

2. Population growth forces agricultural expansion in project area:

Although population is growing in the area, it is clear from the PRA and the LULC change analysis that small-scale agriculture is not a significant driver of deforestation in the area. Nevertheless, the Project includes capacity building on agroforestry techniques to help riverine people and traditional rural villagers to develop more efficient crop systems that require less area and longer rotation times, thus reducing the need of clearing forest patches under regeneration. 50 members of the community have been benefited from such trainings during this monitoring period.

3. Loss of carbon stocks through fire, illegal felling, and land clearing:

The Project has reduced the risk of leakage, illegal logging, and fire by building strong partnerships with the households in the Project Boundaries and its vicinity thus preventing deforestation activities to start. This includes by giving out cook stoves and paying for the riverine and traditional rural villagers in these locations to gain title. Also, capacity building workshops were held to better monitor the forest, thus preventing further LULC change in the area.

Illegal logging risks were mitigated through several measures including demarcating boundaries and posting signage, blocking machinery access through trenching and other methods.

4. The hair comb effect has come very close to the project and the project every day is focused on stopping forest damage.

2.2.7 Benefit Permanence (G1.11)

Community activities are designed to transform local economies over the life of the Project. In this regard, the focus of the project developed local business and income-generating activities that are critical components of a long-term low carbon economy. Project activities that met this overarching objective focus on education, sustainable agriculture, community-based ecotourism, and sustainable management of natural resources. These activities reduced the necessity of community members to deforest and degrade the Project Area. During the Project Lifetime, this is achieved, for example, facilitating better education, through training farmers in sustainable agriculture.

The project focuses on three principal strategies to ensure the maintenance and enhancement of the project benefits beyond the project lifetime. For this monitoring period the focus is the same.

1. Skill and capacity development;
2. Goal of permanent Land ownership;
3. Health benefits.

1. The skills are learnt by the communities throughout the project's lifetime. These relate to better land resource management. The project has initiated several awareness programs for efficient use of land for agricultural practices and has also provided cook stoves which have the benefit of lessening the time for Farinha production and the overall cooking time. The protection of the forests itself ensure that due to lesser degradation there is greater potential to provide timber and non-timber forest products on a sustainable basis. The community has been made aware and trained in alternative crops of agroforestry such as black pepper, honey or andiroba oil. Around 200 community members have received training during this monitoring period.

2. The goal of permanent land ownership to the communities is one of the main initiatives of the project and this provides permanent ownership even beyond the project lifetime. This provides the community to implement the skills and learnings on their own land which is self-sustainable and provides benefits beyond the project's lifetime. Around 127 CARs have been distributed during this monitoring period.

3. The health benefits to the women and to the overall community is expected to continue beyond the project's lifetime. In a 2002 report, WHO listed indoor smoke from solid fuels among the top 10 risks to human health. "Day in and day out, and for hours at a time, women and their small children breathe in amounts of smoke equivalent to consuming two packs of cigarettes per day," WHO reported in the 2006 report Fuel for Life: Household Energy and Health. As greenhouse gas emissions have increased, the smoke from kitchens in the developing world has escalated from a local to a worldwide threat. The average cooking fire produces about as much carbon dioxide as a car, and produces more soot, also known as black carbon. Reducing these emissions may be among the fastest, cheapest ways to fight global climate change. Around 150 women have received cook stoves which translates to improved livelihood during this monitoring period.

The permanence of the benefits associated with the project are captured during the feedback from the community during the periodic community meetings in which around 78% of them prefer to continue the good practices.

2.3 Stakeholder Engagement

2.3.1 Stakeholder Access to Project Documents (G3.1)

All the documents/results are published in the project website of the Syndicate and communicated in Portuguese in a simple language format to the council of stakeholders for their awareness and free participation. The project CCB implementation from validation onwards is described in the relevant section in the MR.

The documents are also going to be posted in the future on the information board of each school being built.

Apart from the survey teams, the main form of communication followed was one-to-one meetings with the community in which the feedback on the benefits provided were communicated periodically. Further during this period, the benefits were mainly related to Skill and capacity development, Goal of permanent Land ownership and improved land management practices which were categorized as both long-term and short-term benefits in the relevant sections of the MR. Samples of the survey forms and attendance records of the one-to-one meetings have been submitted to the VVB.

2.3.2 Dissemination of Summary Project Documents (G3.1)

All the documents/results are published in the project website and communicated in Portuguese in a simple language format to the council of stakeholders for their awareness and free participation.

Many ribeirinhos come to the office each day and ask questions and gain information from the Syndicate and the staff in the office.

Apart from the survey teams, the main form of communication followed was one-to-one meetings with the community in which the feedback on the benefits provided were communicated which are periodically. Further during this period, the benefits were mainly related to Skill and capacity development, Goal of permanent Land ownership and improved land management practices which were categorized as both long-term and short-term benefits in the relevant sections of the MR. Samples of the survey forms and attendance records of one-to-one meetings have been submitted to the VVB

2.3.3 Informational Meetings with Stakeholders (G3.1)

The meetings with the local stakeholders and communities were organized by the PP and the process was similar to the meetings that already took place with them previously (at the beginning of the Project). Each person responsible for every stakeholder group took care to advertise all members of the group about the possibility to participate at the meetings usually through the internal bulletin boards and with e-mails.

The meetings with communities were organized through the following mechanisms:

- At the beginning of each meeting, participants received a summary sheet of the project.
- During the meeting, aspects related to the project and summary information on monitoring results were released and explained.

- There were question and answer sessions after the talks. The questions of the participants were resolved, and all observations were heard and taken into consideration.
- The information provided included contacts (phone number and email) of the people in charge of the project documentation (PP), in order to give the attendants the possibility to permanently communicate their concerns or comments.

2.3.4 Community Costs, Risks, and Benefits (G3.2)

Information about Community Costs, Risks, and Benefits was exchanged and discussed during consultative meetings. The project has no expense to the communities, the team of technicians have always based their intention as go-to-the point; thus the project team goes to the community, we don't request the community to come to the project team. The travel cost is the greatest burden for the community. It was discussed and explained during all meetings that the carbon credit project costs the community nothing, and that the project is 100% there to benefit the locals in a business-as-usual situation.

As explained in the community meetings and one-on-one discussions there is no risk to the community.

The benefits are explained to be cook stoves, land tenure documentation and survey work and training for a better sustainable living and improved livelihoods.

2.3.5 Information to Stakeholder on Verification Process (G3.3)

The Project's executive summary, including project information and project benefits has been translated into Portuguese and is posted in public places in communities throughout the Project Zone.

During community meetings held by project staff as part of the project outreach process, the monitoring and verification process was described.

A poster/notice in Portuguese advertising the Project's public comment period and the validation /verification field visit was posted in communities throughout the Project Zone.

2.3.6 Site Visit Information and Opportunities to Communicate with Auditor (G3.3)

The Project Proponent and Syndicate of Rural Producers has actively communicated to community members and stakeholders about the start of the Public Comment Period and the methods by which they can submit comments on the Project as well as how to view full project documentation. This was accomplished by communicating the Project's Public Comment Period and Verification field visit dates to previously identified stakeholders, community leaders, leaders of the faith communities and public officials. They were then requested to pass that information onto their communities.

Six weeks prior to the site visit in June 2021, verbal communication took place, and it was requested that up to 1 person from all nearby families be prepared to answer questions from the auditor. One-on-one interviews were arranged.

The Syndicate of Rural Producers has a WhatsApp Group that was set up to communicate with the population in a fast way. The Syndicate provided a message to about 400 of the 875 families informing them of the audit. About 475 families do not have phone access.

2.3.7 Stakeholder Consultation (G3.4)

Through local consultation, we assessed the possible impacts of the Project and the perceptions of stakeholders were identified.

So far, the design and implementation of the Project have not been modified, as comments received do not affect the design of the Project. The plan to maintain continuous communication with stakeholders includes a communication channel that addresses possible suggestions and complaints, training activities, dissemination of monitoring reports achievements, etc.

The stakeholders have given inputs on how the social activities are conducted, and have felt the project execution is being completed properly.

2.3.8 Continued Consultation and Adaptive Management (G3.4)

The Syndicate of Rural Workers, one of the main project participants, has the contact information for every family involved in the project.

They know, first-hand, everyone in the community and this knowledge allows for seamless communication between the project team and the local communities.

Throughout the lifetime of the project, we maintain a direct line of communication with community members and relevant stakeholders. This establishes a commitment to communication and consultation to keep stakeholders informed of project activities including restoration, maintenance, monitoring and the CCB validation and verification process.

With-project scenario there is an increase to the socio-ecological resilience, reduced vulnerability and improved adaptation capacity through a better management of the natural resources, including adaptive management. In addition to this, forest protection in the project area does provide a healthy ecosystem with much greater adaptation potential to climate change, with a higher resistance and recovery capability to extreme meteorological phenomena and a wide range of benefits to the neighboring people.

Hence, there is no change in the consultative and adaptive management system during this monitoring period.

2.3.9 Stakeholder Consultation Channels (G3.5)

The stakeholders identified were invited to report and comment in reference to the Project implementation through local consultation. As described in the 2.3.1, we organized consultation meetings with different communities and stakeholders (owners of neighboring farms, stakeholders of the afforestation supply chain, forest technicians, permanent/semi-permanent workers of the farm, Portel and project workers) in the Portel region. The invitation to these events was always conducted by telephone. In addition, various governmental and non-governmental organizations were involved, which we can call “institutional stakeholders”.

The meetings with these institutional stakeholders were organized by means of personal visits. During all the consultation processes the aspects of climate change and carbon markets were addressed by providing information and general concepts in a simple language format that could be understood by all participants. The documentation and information regarding the Project were made available to the community through the following mechanisms:

- At the beginning of each meeting, participants received a summary sheet of the Project for them to understand the Project.

- During the meetings aspects related to forest carbon projects, specific project activities and participants were explained.
- There were question and answer sessions after the talks. The questions of the participants were resolved, and all observations were heard and taken into consideration.
- The information provided, included contact information (phone number and email) of the people in charge of the Project documentation (Project developers), in order to give the attendants, the possibility to permanently communicate their concerns or comments.

2.3.10 Stakeholder Participation in Decision-Making and Implementation (G3.6)

During this monitoring period, consultations have ensured to engage with both men and women, and more marginal stakeholder groups in culturally appropriate ways to ensure that the project can hear a wide range of perspectives. Apart from the survey teams the main form of communication followed was one-to-one meetings with the community.

2.3.11 Anti-Discrimination Assurance (G3.7)

The Ribeirinho REDD+ project has company policies to prevent discrimination and outline a course of action, should it occur, the human resource (HR) policy provides a clear statement on discrimination relating to gender, religion, or sexual discrimination. Discrimination is considered a level A misconduct under the HR policy. Where discrimination occurs within the company, partner organizations, or within project areas (project participants), actions are outlined in the grievance policy to ensure that any discrimination is dealt with by the senior management. All company employees and field partners sign a code of conduct with ES that includes anti-discrimination. No such related incidents have been reported for this monitoring period.

2.3.12 Grievances (G3.8)

The conflict resolution approach sequentially adopted, responds to the conformity or inconformity of the complainant to the proposed solution. The evaluator may also propose a specific approach for the resolution depending on the complexity of the case and the assessment of the same. All the issues are addressed as per the company grievance policy.

The present mechanism does not exclude the right of local people to present the case to any public entities. Actually, during the census, the Project's management team informs local people of the creation of such committee and is informed of their right to present grievances directly to the team on the ground, making a clear point that all claims and/or complaints are addressed equally *notwithstanding the line of grievance*. No such related incidents have been reported for this monitoring period.

The Syndicate of Rural Producers keeps a notebook in the office for grievances related to the project and none have occurred.

The project does not foresee grievances from the local population, besides questions on how fast it will be done, so the environmental payments can start.

The land certificates are in demand, now that 1645 families in the region have gained these, more families have come to the office every day seeking this certificate.

2.3.13 Worker Training (G3.9)

Project activity #2 (after Plots for the Poor) is designed to provide training to local community groups (in jatai, honey, and black pepper production activities) that generates the required capabilities to undertake forest monitoring as well as monitoring for social and biodiversity variables.

The Project ensures that all members of the community groups have the same opportunity to attend capacity building workshops and participate in demonstrational activities, regardless of race, religion, sexual orientation, or gender. Although the opportunity to actively participate in all the activities of the Project is present, in the end it is the decision of each stakeholder to participate, with no pressure.

Special attention was given to under-represented groups (elder people, woman, and children) to be aware of the on-going training workshops and activities. The content and language of capacity training and demonstrational activities was adapted accordingly for each participating group. The Project considers developing and implementing workshops specifically designed for these age classes groups in order to better transmit the information.

Due to the existing educational level within the under-represented groups in the region, the one-on-one workshops took adequate time in explaining how things could be done differently. Sometimes the training and explanations had to define even common words; i.e., what is “carbon”, what is a “credit”. Both words are new for most of the people in the community. The project did its best to help bring a different insight, but most community groups wanted to focus on what they knew already which is Cassava.

Capacity building is a relevant aspect in the implementation and operation of the project. In order to achieve the goals successfully, the members of the community must be trained to have the skills and knowledge to effectively carry out the work.

50 community people have benefited from the training programs during this monitoring period.

To fulfil their responsibilities, the recruiting manager is responsible for integrating and managing confidential personnel information, verifying information provided, and drawing up contracts. Once hired, the staff goes through a trial period. For the selection of officials, the human resources team will have the objective to find qualified and reliable staff whose skills are in line with the requirements and objectives of the company, through technical, transparent, and non-discriminatory procedures, based on merits and excellence.

Brazil has horrendous labor laws so each employee has to sign the “Registro de Ponto” and this basically requires them to attest to their hours they work.

2.3.14 Community Employment Opportunities (G3.10)

The Project designed employment opportunities to make sure underrepresented groups of the local community have equal opportunities of finding employment within the Project management and demonstrative activities. Recruitment was done as per the recruitment policy of the company.

Employment positions that require demanding physical work and a higher risk (i.e., on the ground monitoring of former logging trails, sampling biomass in forest plots, monitoring of Project Boundaries by boat to detect illegal logging activities, setting and revisiting biodiversity camera traps) are filled by persons between the age of 18 and 60 years and/or according to the experience and physical strength of a person, assessed on an individual basis.

Other employment opportunities that require less physical effort and by their nature are less risky are for the elderly people or less physically apt people. Such activities can be, but are not limited to: social assessment surveying and monitoring, running demonstrative activities to other members of the local communities and to neighbor communities, and actively participating in setting up demonstrative activities (i.e., home gardens, forest gardens, improved fallows, energy efficient cook stoves, etc.).

The project has over 63 people working in total, of which 7 are woman. With a goal to ramp up to over 300 people working upon approval of the Ribeirinho REDD project. Many of the jobs are construction

related jobs building schools and woman are not interested in this work. The mass employment is due to the number of schools and teachers that will be needed.

2.3.15 Relevant Laws and Regulations Related to Worker's Rights (G3.11)

The local community received clear and adequate information about the requirements of national and international regulations on worker's rights before entering in a contract agreement with the Sindicato for the work being conducted.

The Sindicato made sure to comply with the applicable national regulations on worker's rights. This is assured by yearly audits held by a third party that was identified when the census was done. Such audits were announced to local authorities and communities, and they were encouraged to meet with the audit entity. This way, local people can rest assure that all their complaints about worker's rights are known in a straightforward and clear way.

The following is a list of Brazil's all relevant laws and regulations covering worker's rights:

The Brazilian Constitution, Chapter II-Social Rights, Articles 7-11 which addressed: Minimum wage, Normal working hours, Guidance on vacation and weekly leave, Guidance on maternity and paternity leave, Recognition of collective bargaining, and Prohibition of discrimination.

In addition to the Constitution, there are two additional decrees related to Brazilian labor laws. Consolidação das Leis do Trabalho (CLT): DECRETO-LEI N.º 5.452, DE 1º DE MAIO DE 1943 (Consolidate of Working Laws). This decree gives more clarification on: Hourly, daily, weekly and monthly work hours of Employment of minors and women, Establishes a minimum wage of Worker safety and safe working environments, Defines penalties for non-compliance by employers, and Establishes a judicial work-related process for addressing all worker related issues.

Estatui normas reguladoras do trabalho rural: LEI No 5.889, DE 8 DE JUNHO DE 1973 (Establishes Regular Norms for Rural Workers). This is a complimentary law to the aforementioned 1943 decree because prior to 1973, rural workers did not have the same rights as urban workers. In 1973, this law was established to specify the equality between urban and rural workers, along with compensation for overtime.

Thus, it is stated that the project complies with relevant host country rules for this monitoring period as there have been no reported incidents to the contrary.

Other laws related to workers:

- Lei 605/1949 - Repouso Semanal Remunerado (Paid Weekly Rest);
- Lei 2.959/1956 - Contrato por Obra ou Serviço Certo (Contract for Work or Right Service);
- Lei 3.030/1956 - Desconto por Fornecimento de Alimentação (Discount for Food Supply);
- Lei 4.090/1962 - Gratificação de Natal (Christmas Bonus);
- Lei 4.749/1965 - 13º Salário (13th salary);
- Lei 4.886/1965 - Representantes Comerciais Autônomos (Autonomous Business Representatives);
- Lei 4.950-A/1966 - Remuneração de Profissionais (Engenharia, Química, Agron. e Veter.) (Remuneration of Professionals (Engineering, Chemistry, Agron. And Veter.);
- Lei 5.859/1972 - Empregado Doméstico (Housekeeper);
- Lei 5.889/1973 - Trabalho Rural (Rural Work);
- Lei 6.019/1974 - Trabalho Temporário Urbano (Temporary Urban Work);
- Lei 6.494/1977 - Estagiários (Trainees);
- Lei 6.919/1981 - FGTS de Diretores (FGTS of Directors);
- Lei 6.932/1981 - Médicos Residentes (Resident Doctors);
- Lei 7.418/1985 - Vale-Transporte (Transportation vouchers);
- Lei 8.036/1990 - Lei do FGTS (FGTS Law);
- Lei 8.906/1994 - Advogados (Lawyers);

- Lei 9.601/1998 - Banco de Horas e Contrato por Prazo Determinado (Bank of Hours and Contract for Term Determined);
- Lei 10.101/2000 - Participação dos Trabalhadores nos Lucros ou Resultados (Workers' Participation in Profits or Results);
- Lei 10.607/2002 - Declara Feriados Nacionais (National Holidays);
- Lei 10.748/2003 - Programa Primeiro Emprego – PNPE (First Job Program);
- Lei 10.820/2003 - Desconto de Prestações em Folha de Pagamento (Discount on Payroll Benefits);

2.3.16 Occupational Safety Assessment (G3.12)

Project's activities do not hold risk besides those inherent to the day-to-day life in the forest. Project's activities do not require the use of heavy machinery or dangerous substances. Nevertheless, the Project management team has been provided adequate protection equipment to employees working in forest monitoring activities. Also, monitoring staff were equipped with first aid kits. Protection equipment included but was not limited to:

Hard hat
Cap with the company's logo
Reflective/fluorescent security vest with the company's logo
Rubber boots
Gloves
Two-way radios
GPS
Camera
Field backpack
Camel pack
LED Flashlight
Whistle
Machete
First aid kit

Risks for each type of work was assessed and safety guidelines were developed helping identifying and reducing such risks. Guidelines were written in clear and adequate language and distributed among workers. Additionally, workers have received safety instructions to make sure any doubts and suggestions are taken care of.

PP has a safety policy that is shared with employees and is on the PP's Mobile website.

During this period such training was provided to 20 people who have been employed of which 3 are women.

2.4 Management Capacity

2.4.1 Required Technical Skills (G4.2)

- Construction Skills to build schools
- Organization Skills to coordinate a large number of staff
- Relationship skills to build the relationships that this project has done with SEMA, DEMA, Doctors without Borders, and UNICEF, as well as the Mayor's Office.
- Technical skills related to carbon Calculations
- Social Works and Anthropology skills to work with the Ribeirinho population.

2.4.2 Management Team Experience (G4.2)

Dr. Laura Carolina Vieira, doctorate in Anthropology, Post Graduate from the University Federal of Para. Bachelors in History from the Federal University of Parana. Member of the Research Group of Anthropology and Tourism for the Amazon (GATA). The author of the following papers on anthropology: 1.) VIEIRA, L. C.. The Erveiras women of Ver-O-Peso and heritage views. PATHS OF HISTORY UNIMONTES) (ONLINE), v. 24, p. 97, 2020. 2.) VIEIRA, L. C.. First steps in the herbal sector. Four Fields Notebook, vol. 1, p. 35-38, 2019. 3.) VIEIRA, LC The Varied Agencies of the Herb Sector of the Ver-o-Peso Fair and the Hegemonic Urban Context: commodification of cultural goods in gentrification processes in the Eastern Amazon.. In: XIII RAM - Anthropology Meeting of Mercosur, 2019, Porto Alegre. RAM 2019, 2019. 4.) VIEIRA, L. C.. Correspondences and Dynamics between the Herb Sector in the Ver-o-Peso Market and the Urban Context. In: 31st RBA - Brazilian Meeting of Anthropology, 2018, Brasília. Proceedings of the 31st RBA, 2018. 5.) VIEIRA, L. C.. Tourism and the Smell Bath: Mechanisms Used in the Ver-o-Peso Fair for the Composition of Cultural Imagery in Belém, Pará. In: V Transborders, 2020, Manaus, Tefé. Transborder, 2020. 6.) VIEIRA, L. C.. Knowledge of the Forest, Products in the City? Rebuttals of the herb culture as a traditional Amazonian practice in relation to metropolitan socioeconomic projects. In: IV Ecuadorian Anthropology Meeting - Diversities, Adversities, Resistances., 2019, Salvador. IV REA, 2019. 7.) VIEIRA, L. C.. Tourism and the Smell Bath: Mechanisms Used in the Ver-o-Peso Fair for the Composition of Imagery Cultural de Belém, Pará. 2020. (Presentation of Work/Communication). 8.) VIEIRA, L. C.. The Photogenic Posing of Erveiras in Belém do Pará: strategy and self-esteem for Ver-o-Peso stallholders. 2020. (Presentation of Work/Congress). 9.) VIEIRA, L. C.. The Erveiras do Ver-O-Peso women and heritage views. 2020. (Presentation of Work/Communication). 10.) VIEIRA, L. C.. The Varied Agencies of the Herb Sector of the Ver-o-Peso Fair and the Hegemonic Urban Context: commodification of cultural goods in gentrification processes in the Eastern Amazon. 2019. (Presentation of Work/Communication). 11.) VIEIRA, L. C.. Knowledge of the Forest, Products in the City. Rebuttals of herb culture as a traditional Amazonian practice in relation to metropolitan socioeconomic projects.. 2019. (Presentation of Work/Congress). 12.) VIEIRA, L. C.. In the light of Ethnoecology: The Herb Market and its materials as sensory powers for perceptions, representations and identities. 2018. (Submission of Work/Other). 13.) VIEIRA, L. C.. Correspondences and Dynamics between the Herb Sector in the Ver-o-Peso Market with the Urban Context.. 2018. (Presentation of Paper/Congress). 14.) VIEIRA, L.C.; OLIVEIRA, K. . Curatorship of the Collection of the Textual, Sound and Visual Documentation Unit of MAE. 2016. (Presentation of Work/Congress). The goal I have for the project is to end the poverty situation in Portel via the programs environmental payments.

Ana Paula Soares Farias - Graduated in the course of Social Sciences at the Federal University of Pará (2021). Acting as a scholar in the project of Academic Monitoring (PGRAD-UFPA) entitled "Gender relations and cultural practices in the field of Paraense culture", which enabled the experience in teaching practice. Scientific Initiation Scholar in the years 2018 to 2020 where I developed studies in the research line gender and artisanal fisheries in protected areas of the Amazon in the project " The role of women in fisheries governance and management of fishery resources in the Japurá River region, Amanã Sustainable Development Reserve, Amazonas." (2018-2020). Member of the research group GEPTIGAM- Study Group on Traditional Populations, Identity, Gender and Environment (UFPA). The Ribeirinho REDD Project the focus is equality, gender equality, woman's rights, human advancement, and overseeing the technicians for the Acai and honey bee project to make sure it fulfills all the requirements set out by VERRA.

Dr. Paula Tavares, a veterinarian with a focus on wild animals. At the beginning of my graduation, completed an internship at a veterinary hospital that only attended to conventional pets (dogs and cats) but it was not exactly my life's goal, so I went looking for an internship in other places, until I was welcomed at Jardim Bosque Rodrigues Alves, there I had contact with the local wildlife and felt at home, during this period I participated in some projects and one of them was the monitoring of sea turtles on the beaches of Curuçá. After 2 years at Bosque Rodrigues Alves, I decided to go out and try something new, so I volunteered for the Environmental Police Battalion for about a year, I even won a decoration. At the end of graduation I went to southern Brazil (Santa Catarina) to do my mandatory internship in the Beach Monitoring Project - PMP. There I had contact with marine animals, most were birds (birds are my biggest passion)

but there were also some dolphins and turtles. Now that I'm graduated, my main area of expertise is Fauna Rescue, which I still practice today. Author: Academic work "Environmental Enrichment Tool to Improve the Well-Being of Manatees (*Trichechus inunguis*, LINNAEUS, 1758) In Captivity at the Bosque Rodrigues Alves Zoo and Botanical Gardens in Belém do Pará, Brazil". No II Congress of Veterinary Doctors of the Legal Amazon - Amazonvet / I National Congress of Wild Animals, atbanner category, from August 20th to 22nd, at Hangar - Amazon Convention and Fair Center, Belém / To Brazil. Co-Author: Academic work on "Hemipenectomy in Iguana iguana". Featured in the banner category, at the 41st Physician Week Veterinary - Semavet / 9th Paraense Symposium on Veterinary Medicine. On September 4th and 5th, 2017, at the Hotel Sagres, in Belém / Pará, Brazil. Co-author: Academic work "Effect of Environmental Enrichment on the Well-Being of the Smell Monkey (*Saimiri Sciureus*, LINNAEUS, 1758) In Captivity at the Bosque Rodrigues Alves Zoo and Botanical Garden in Belém / Pará, Brazil". Displayed in the banner category at the II Congress of Veterinary Doctors of the Legal Amazon - Amazonvet / I National Congress of Wild Animals, held from August 20 to 22, at Hangar - Amazon Convention and Fair Center, Belém / Pará, Brazil. Co-author: Academic work "Analysis of the Environmental Enrichment Technique for Ocelot (*Leopardus pardalis*, LINNAEUS, 1758) In Captivity at the Bosque Rodrigues Alves Zoo and Botanical Garden in Belém do Pará, Brazil". Displayed in the banner category at the II Congress of Veterinary Doctors of the Legal Amazon - Amazonvet / I National Congress of Wild Animals, held from August 20 to 22, at Hangar - Amazon Convention and Fair Center, Belém / Pará, Brazil. The Ribeirinho REDD project is a perfect REDD project in a region that is at incredibly high risk of full conversion and the programs I am conducting for the project will both save numerous endangered species but also train the locals to monitor their own activities to protect those species.

2.4.3 Project Management Partnerships/Team Development (G4.2)

Dr. Paresh Patel - hired by the Sindicato of Rural Producers to prepare the PD, MR, PD in local language and MR in local language. Dr. Paresh Patel has consulted on the writing and calculations of numerous REDD carbon credit projects in the past. His expertise comes from more than 10 years as a forestry engineer, having a PhD in Environmental Sciences. Dr. Paresh Patel's work is limited in the VERRA REDD field, as purely a document preparation person. Dr. Paresh Patel's expertise has been accumulated over the years working in the Democratic Republic of the Congo, monitoring forest timber extraction and FSC forestry in a sustainable manner. Dr. Paresh has also worked in Bolivia, for a charity hoping to help develop a system to track sustainable logs from the farm to the mill. Dr. Paresh has worked for 2 years in Borneo studying the impact of illegal logging on the local indigenous population and who their livelihoods have been altered.

2.4.4 Financial Health of Implementing Organization(s) (G4.3)

The project is funded by the landowners. No outside investors are involved in the Ribeirinho REDD+ project.

The financial mechanisms adopted is the following:

- Forestry activities (including feasibility study, site preparation, seedling, planting, cutting test, maintenance, etc.) were funded with private resources by the Project Owner only.
- Project costs (including Project Design Document, validation, verification, registration, etc.) were funded by Project Proponent only.

This breakdown of costs gives to the Project Owner the opportunity to concentrate his financial efforts only on the implementation of the forest activities and related co-benefits.

The financial mechanisms adopted to achieve the Project climate, community and biodiversity benefits is generated by the following revenue streaming:

- Mainly because of the revenue generation from the selling of carbon credits.

Project Owner has taken all the responsibility to cover all the costs and has been useful for the maintenance of project area. Based on the carbon credit sales next verification and developments will be focused. Project implementation and continuation of achieving CCB benefits is mainly dependent on carbon credit sales, since this is the only revenue generation stream for REDD projects globally.

2.4.5 Avoidance of Corruption and Other Unethical Behavior (G4.3)

The Syndicate working with other groups, helped the riverine gain their land tenure certificates in order to allow them to do this REDD project.

The project is not obligatory, and no one was forced into the project. About 10% of the riverine did not accept, and thus, the locations where their communities are, were excluded.

2.4.6 Commercially Sensitive Information (Rules 3.5.13 – 3.5.14)

There is no commercially sensitive information in this monitoring report document, itself. Supporting documents which include commercially sensitive information that are not made publicly available include: the MOU; Contracts with Buyers and Service Providers; and documents related to project financials.

2.5 Legal Status and Property Rights

2.5.1 Recognition of Property Rights (G5.1)

Name of the Law	Description	Project Compliance
Law number 4771, September 15th 1965 (D.O.U of September 16th 1965)	The Brazilian forest code of 1965 – Brazilian Forest Code –provides for example: II – area of permanent preservation: protected area in the terms of article 20 and 30 of this law, covered or not by native vegetation, with the role of protecting the water resources, landscape, geological stability, biodiversity, flux of genes of plants and animals, protect de soil and secure a good environment for the human population; III – Legal Reserve Areas: Area located in the property or “posse rural” excluding the areas of permanent preservation, for the sustainable use of the natural resources, conservation and restoration of the ecological process, biodiversity conservation and refugee and protection of native animals and plants; Art. 150 – It is prohibited under empirical form the exploration of primitive forest of the Amazon watershed, but only can be explored in accordance of technical management plans approved by act of Public authorities, to be issued in one year <i>term</i> .	All properties have legal reserve areas and APPs defined. In accordance with the CARs (Environment Rural Registry) at SEMA (Environment State Institute)
Normative Instruction	Regulatory of the Environmental Rural Registry -CAR in the state of Pará and providence of other requirements. Art 1 – establish criteria and procedures for	Development of CAR in all lands in the

number 003 of May 23th 2007 – Executive office of environment , science and technology - SECTAM	implementation of the CAR – PA as an instrument for identification of the rural properties in the state of Pará that must be issued by SECTAM-PA in accordance with this Normative Instruction. Art 2 – It is necessary for all rural properties in the state of Pará to be registered in the CAR-PA, even the properties that have no production activity. Art 3 – The issuance of the CAR-PA, as toll for identification of the property was done only once for each property. It has a registry number with a sequential number. This number is in all licenses, authorizations, and other documents issued for the environmental regularization of the rural property. This registry number is linked to the land, independent if the land is sold, transferred or taken possession by another person. Single paragraph – There are no concessions of any license for the land that has no registry at CAR-PA. Art 4 – In the CAR-PA it is mentioned all the basic data of the rural property, Total area- AT, Area of permanent preservation – APP, legal reserve areas – ARL, and area of alternative use of the soil – AUAS, in addition the name and profession of the landowner, geographic coordinates and other information required by complementary laws	Project Area
Federal Decree number 5.975/2006	Art. 10 – Forest exploration and succession formations that require shallow harvest of the forest only are permitted under specific authorization for alternative land uses issued by SISNAMA. # 1o By alternative land use is understood any conversion of the forest to other land cover, such as settlements, agriculture, pasture, industry, energy generation, mining and transportation.	All properties have legal reserve areas and APPs defined. In accordance with the CARs (Environment Rural Registry) at SEMA (Environment State Institute)

2.5.2 Free, Prior and Informed Consent (G5.2)

The tenancy agreements between the the Syndicate of Rural Producers and the Riverine is done in a legal and clear manner.

2.5.3 Property Right Protection (G5.3)

Our project activities do not lead to involuntary removal or relocation of property rights holders from their lands or territories and does not force rights holders to relocate activities important to their culture or livelihood.

There will never be a relocation of anyone unless they decide to relocate. If in the future any relocation of activities needs to be undertaken, it will take place with a written agreement that demonstrates that the agreement was made with the free, prior, and informed consent of those concerned and includes provisions for just and fair compensation.

All residents, even land grabbers, are allowed to keep the land they have taken up to 100 hectares.

However, land grabbing is not tolerated, and when it occurs inside the project area, where no one has been living since the project started in 2017, it will not be protected.

2.5.4 Identification of Illegal Activity (G5.4)

Illegal activities in the area are constituted by unplanned timber extraction. Such logging operations are evidenced by the proliferation of pioneer roads as presented. It is known from literature that extractive operations take advantage from the fact that local farmers don't have land titles to displace them or to gain access to the forest resources in nearby villages (Araujo, Bonjean et al. 2009). At the same time, illegal logging operations thrive whenever there are forested areas that seem to be under no-use and where the presence of the landowner is not made evident (Margulis 2004).

The Project has trained local community members to work as monitoring staff in the Project Area and the LMA. This is their main activity, to identify, prevent, and avoid illegal activities from taking place in the Project Area.

Electronic monitoring of government databases allows the project to monitor the area and different land claims someone may be trying to make.

As support measures against illegal activities, the Project did provide land titles against conservation results to households living within the Project LMA Boundaries and did provide support to neighboring communities to achieve land tenure on unused public lands.

Stakeholders in neighboring communities were encouraged to report encroachers and illegal loggers trying to get into nearby forests. The Project did proceed to make the respective denounce to local authorities just as if the situation was occurring in the Project Area. Through this mechanism the project was generating positive leakage.

No identification of illegal activities was reported for this monitoring period.

2.5.5 Ongoing Disputes (G5.5)

The illegal loggers are against the REDD project. The REDD projects in Portel have greatly deprived the illegal loggers of wood resources and has stopped many illegal logging barges.

2.5.6 National and Local Laws (G5.6)

Even though there aren't any national law or regulations regarding REDD policies, there are some local initiatives to encourage REDD projects. Most of these initiatives are at the municipality level. For example: Paragominas, a municipality located at about 400 km from the project area, has approved (July 26th, 2011) a municipality environmental policy (Law number 765/2011) that includes REDD. With this it was created a municipal-level system for reduction of emissions from degradation and deforestation that will be linked to a potential national or state REDD system.

As of July 1, 2019, the following is a list of all the international, national and state-level laws and regulations identified by the Project Proponents which are relevant to the Project. To the best of the Project Proponents' knowledge, there have not been any new laws relevant to the Project since the Project's inception.

Brazil is a party to numerous international conventions and treaties, such as:

1. Convention on Biological Diversity;
2. United Nations Framework Convention on Climate Change;
3. Convention on International Trade in Endangered Species of Wild Fauna and Flora;
4. International Tropical Timber Organization (i.e., Brazil is a Producing Member);

5. Ramsar Convention on Wetlands;
6. Universal Declaration of Human Rights;
7. United Nations Declaration on the Rights of Indigenous Peoples;
8. Convention on the Elimination of All Forms of Discrimination Against Women; and
9. International Labor Organization Convention.

There was also a Memorandum of Understanding (MOU) signed on March 3, 2010, between Brazil and the United States of America on “cooperation regarding climate change.

Furthermore, there was an international MOU between California (United States), Chiapas (Mexico) and Acre (Brazil) signed on November 16, 2010.

Local Legislation

The main laws governing the forestry sector in the State of Pará are:

- State Law no. 7,389, of 04/01/2012: Defines the activities of local environmental impact in the State of Pará and other measures.
- State Law no. 7,381, of 3/19/2010: Provides for the restoration of the vegetation cover, of the riparian forests of the State of Pará.
- State Law no. 6,745, of 6/6/2005: Establishes the Ecological-Economic Macro zoning of the State of Pará and other measures.
- State Law no. 6,506 of 12/02/2002: It establishes the basic guidelines for the realization of the Ecological-Economic Zoning (EEZ) in the State of Pará and other measures.
- State Law no. 6,462, of 7/4/2002: Provides for the State Policy on Forests and other forms of vegetation.
- State Law no. 5,977, of 7/10/1996: Provides for the protection of wildlife in the State of Pará.
- State Law no. 5,887, of 5/9/1995: Provides for the State Environmental Policy and other measures.
- State Decree no. 518, of 09/05/2012: Establishes the Para-Forum of Climate Change and other measures.
- State Decree no. 216, of 9/22/2011: Provides for the environmental licensing of agro-silvopastoral activities carried out in altered and/or underutilized areas outside the legal reserve area and permanent preservation area in the rural properties of the State of Pará.
 - State Decree no. 2,436, of 8/11/2010: Regulates the actions related, directly or indirectly, to agro-silvopastoral activities, carried out within the areas of alternative land use, considered to be of low environmental impact.
- State Decree no. 2,099, of 1/27/2010: It provides for the maintenance, decomposition, conduction of natural regeneration, compensation, and composition of the Legal Reserve area of rural properties in the State of Pará and other measures.
- State Decree no. 1,697, of 6/5/2009: Establishes the Prevention, Control and Alternatives Plan for the deforestation of the State of Pará and other measures.
- State Decree no. 1,148, of 7/17/2008: Provides for the Rural Environmental Registry - CAR-PA, Legal Reserve area and other measures.

- State Decree no. 58, of 11/27/2006: Establishes the Register of Explorers and Consumers of Forest Products of the State of Pará - CEPROF-PA and the System of Commercialization and Transportation of Forest Products of the state of Pará SISFLORA-PA and its operational documents and other measures.
- State Decree no. 56, of 3/31/2006: Regulates provisions of State Law No. 6,462 of July 4, 2002; which provides for the State Policy on Forests and Other Forms of Vegetation and provides other measures, aiming at encouraging the recovery of altered and/or degraded areas and restoring legal reserve, for energy, wood, fruit, industrial or other purposes, through reforestation and agroforestry with native and exotic species and other measures. - State Decree no. 856, of 01/30/2004: Regulates the Register of Forest Activity.
- Resolution no. 54, of 10/24/2007, Homologates the list of endangered species of flora and fauna in the State of Pará.

The Ribeirinho REDD+ Project complies with all the law requirements of land use, not affecting natural forests and strategic ecosystems.

3 CLIMATE

3.1 Monitoring GHG Emission Reductions and Removals

3.1.1 Data and Parameters Available at Validation

Data / Parameter	Deforestation
Data unit	Hectare (ha)
Description	Maps of forest cover areas converted into non-forest cover areas
Source of data	Measured through data from the PRODES/INPE Project
Value applied	0.40%/year on average (2000-2014)
Justification of choice of data or description of measurement methods and procedures applied	Data from the PRODES Digital program (official mapping satellite of Brazilian Amazon Forest) were used to map the deforestation and production of the Forest Cover Excellence Brand Map. During the analyzed period, a total of 46 Landsat images were used. And for the classification of the images in the mapping of forest classes, non-forest vegetation, hydrography and deforestation, the ISOSEG method of unsupervised classification was used
Purpose of data	- Determination of baseline scenario - Calculation of baseline emissions - Calculation of project emissions - Calculation of leakage
Comments	View the documents: - Câmara et al. 2006. Methodology for the calculation of the annual rate of deforestation in the Legal Amazon - Determination of the Forest Carbon Stock for the REDD+ Project

Data / Parameter	CF
Data unit	T
Description	Carbon contained in dry biomass
Source of data	Nogueira et al. (2008). Estimates of forest biomass in the Brazilian Amazon: New allometric equations and biomass adjustments of wood volume inventories. Forest Ecology and Management, v. 256, n. 11, p. 1853-1867, 2008
Value applied	0.485
Justification of choice of data or description of measurement methods and procedures applied	Value found in scientific literature
Purpose of data	- Determination of baseline scenario - Calculation of baseline emissions - Calculation of project emissions - Calculation of leakage
Comments	

Data / Parameter	ABSLRRt										
Data unit	Ha										
Description	Annual area of baseline deforestation within the RR at year t										
Source of data	Calculated, see Annex VM0015 tables										
Value applied	Table 9a, 11a Annex VM0015 tables <table border="1"> <tr> <td>2017</td><td>18,788</td></tr> <tr> <td>2018</td><td>18,788</td></tr> <tr> <td>2019</td><td>18,788</td></tr> <tr> <td>2020</td><td>6,263</td></tr> <tr> <td></td><td></td></tr> </table>	2017	18,788	2018	18,788	2019	18,788	2020	6,263		
2017	18,788										
2018	18,788										
2019	18,788										
2020	6,263										
Justification of choice of data or description of measurement methods and procedures applied	Calculated according to requirements of the VM0015 v1.1.										
Purpose of data	Calculation of baseline emissions										
Comments	Activity data for calculating GHG emissions in the baseline scenario										

Data / Parameter	ABSLRR
Data unit	Ha
Description	Cumulative area of baseline deforestation in the reference region at year t
Source of data	Calculated, see VCS Annex
Value applied	Table 9a, 11a Annex VM0015 tables 81,415
Justification of choice of data or description of measurement methods and procedures applied	Calculated according to requirements of the VM0015 v1.1.
Purpose of data	Calculation of baseline emissions
Comments	Activity data for calculating GHG emissions in the baseline scenario

Data / Parameter	ABSLPA _t										
Data unit	Ha										
Description	Annual area of baseline deforestation in the project area at year t										
Source of data	Calculated, see VCS Annex ER sheet										
Value applied	Table 9b, 11b, 13b of Annex VM0015 <table border="1"> <tr> <td>2017</td><td>2,248.2</td></tr> <tr> <td>2018</td><td>2,571.6</td></tr> <tr> <td>2019</td><td>2,457.8</td></tr> <tr> <td>2020</td><td>752.0</td></tr> <tr> <td></td><td></td></tr> </table>	2017	2,248.2	2018	2,571.6	2019	2,457.8	2020	752.0		
2017	2,248.2										
2018	2,571.6										
2019	2,457.8										
2020	752.0										
Justification of choice of data or description of measurement methods and procedures applied	Calculated according to requirements of the VM0015 v1.1.										
Purpose of data	Calculation of baseline emissions										
Comments	Activity data for calculating GHG emissions in the baseline scenario										

Data / Parameter	ABSLPA _{icl,t}
Data unit	Ha

Description	Area of initial (pre-deforestation) forest class icl deforested at time t within the project area in the baseline										
Source of data	Calculated, see VCS Annex ER sheet										
Value applied	Table 11b of Annex VM0015 <table border="1"> <tr> <td>2017</td><td>2,248.2</td></tr> <tr> <td>2018</td><td>2,571.6</td></tr> <tr> <td>2019</td><td>2,457.8</td></tr> <tr> <td>2020</td><td>752.0</td></tr> <tr> <td></td><td></td></tr> </table>	2017	2,248.2	2018	2,571.6	2019	2,457.8	2020	752.0		
2017	2,248.2										
2018	2,571.6										
2019	2,457.8										
2020	752.0										
Justification of choice of data or description of measurement methods and procedures applied	Calculated according to requirements of VM0015 v1.1, 5.1 by applying land cover map to the result of Table 9b										
Purpose of data	Calculation of baseline emissions										
Comments	Activity data for calculating GHG emissions in the baseline scenario										

Data / Parameter	ABSLPA _{i,t}				
Data unit	Ha				
Description	Annual area of baseline deforestation within stratum (i) of the project area at year t				
Source of data	Calculated, see VCS Annex				
Value applied	Table 9b of Annex VM0015 <table><tr><td>2,248</td></tr><tr><td>2,572</td></tr><tr><td>2,458</td></tr><tr><td>752</td></tr></table>	2,248	2,572	2,458	752
2,248					
2,572					
2,458					
752					
Justification of choice of data or description of measurement methods and procedures applied	Calculated according to requirements of VM0015 v1.1, 4.1.2.2				
Purpose of data	Calculation of baseline emissions				
Comments	Activity data for calculating GHG emissions in the baseline scenario				

Data / Parameter	ABSLPA
Data unit	Ha

Description	Cumulative area of baseline deforestation within the project area at year t
Source of data	Calculated, see VCS Annex
Value applied	Table 9b, Table 11b, Table 13b of Annex VM0015 9,725
Justification of choice of data or description of measurement methods and procedures applied	Calculated according to requirements of the VM0015 v1.1.
Purpose of data	Calculation of baseline emissions
Comments	Activity data for calculating GHG emissions in the baseline scenario

Data / Parameter	ABSLPAz,t								
Data unit	Ha								
Description	Area of the zone z “deforested” at time t within the project area in the baseline case; ha								
Source of data	Calculated, see VCS Annex ER sheet								
Value applied	Table 13b of Annex VM0015 <table border="1"> <tr> <td>2017</td><td>2,248</td></tr> <tr> <td>2018</td><td>2,572</td></tr> <tr> <td>2019</td><td>2,458</td></tr> <tr> <td>2020</td><td>752</td></tr> </table>	2017	2,248	2018	2,572	2019	2,458	2020	752
2017	2,248								
2018	2,572								
2019	2,458								
2020	752								
Justification of choice of data or description of measurement methods and procedures applied	Equal to values of Table 11b grouped by zones.								
Purpose of data	Calculation of baseline emissions								
Comments	Calculating net GHG emissions via post-deforestation C-stocks								

Data / Parameter	ABSLLKt		
Data unit	Ha		
Description	Annual area of baseline deforestation within the leakage belt at year t		
Source of data	Calculated, see VCS Annex ER sheet		
Value applied	Table 9c, 11c, 13c of Annex VM0015 <table border="1"> <tr> <td>2017</td><td>16,540</td></tr> </table>	2017	16,540
2017	16,540		

	2018	16,216	
	2019	16,330	
	2020	5,511	
Justification of choice of data or description of measurement methods and procedures applied	Calculated according to requirements of VM0015 v1.1.		
Purpose of data	Calculation of leakage		
Comments	Activity data for calculating GHG emissions in the baseline scenario		

Data / Parameter	ABSLLKicl,t									
Data unit	Ha									
Description	Area of initial (post-deforestation) forest class fcl deforested at time t within the leakage belt in the baseline case									
Source of data	Calculated, see VCS Annex ER sheet									
Value applied	Table 11c of Annex VM0015 <table><tr><td>2017</td><td>16,539.7</td></tr><tr><td>2018</td><td>16,216.3</td></tr><tr><td>2019</td><td>16,330.2</td></tr><tr><td>2020</td><td>5,511.0</td></tr></table>		2017	16,539.7	2018	16,216.3	2019	16,330.2	2020	5,511.0
2017	16,539.7									
2018	16,216.3									
2019	16,330.2									
2020	5,511.0									
Justification of choice of data or description of measurement methods and procedures applied	Calculated according to requirements of VM0015 v1.1, 5.1 by applying land cover map to the result of Table 9c									
Purpose of data	Calculation of leakage									
Comments	Activity data for calculating GHG emissions in the baseline scenario									

Data / Parameter	ABSLLKI,t	
Data unit	Ha	
Description	Annual area of deforestation in stratum (i) within the leakage belt at year t	
Source of data	Calculated. See VCS annex – ER sheet	
Value applied	Table 9c, 11c, 13c of Annex VM0015.	
	2016	17,092

	2017	16,540	
	2018	16,216	
	2019	16,330	
Justification of choice of data or description of measurement methods and procedures applied	Activity data for calculating GHG emissions. Calculated according to requirements of VM0015 v1.1.		
Purpose of data	Calculation of leakage		
Comments	Activity data for calculating GHG emissions in the baseline scenario		

Data / Parameter	ABSLK
Data unit	Ha
Description	Cumulative area of baseline deforestation within the leakage belt at year t
Source of data	Calculated
Value applied	Table 9c, 11c, 13c of Annex VM0015. 71,689
Justification of choice of data or description of measurement methods and procedures applied	Calculated according to requirements of VM0015 v1.1.
Purpose of data	Calculation of leakage
Comments	Activity data for calculating GHG emissions in the baseline scenario

Data / Parameter	CFj
Data unit	Dimensionless
Description	Carbon fraction for tree tr, of species, group of species or forest type j
Source of data	IPCC GPG 2006 Chapter 6
Value applied	forest classes: 0.5 Post-deforestation classes: 0.47
Justification of choice of data or description of measurement methods and procedures applied	Default values IPCC GPG 2006, Chapter 6
Purpose of data	Calculation of baseline emission

Comments	Conversion from biomass to CO ₂ e
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Data / Parameter	Cabcl
Data unit	t CO ₂ e ha ⁻¹
Description	Average carbon stock per hectare in the above-ground biomass carbon pool of LU/LC class cl
Source of data	ER sheet
Value applied	forest class: bh-M: 250.75 bh-MB: 350.22 bh-PM: 196.63 bmh-M: 216.28 bmh-MB: 448.24 bmh-PM and and bmh-PMt: 460.92 bms-T: 169.23 bp-M: 183.36 bp-MB: 181.47 bp-PM: 398.95 bs-MB: 165.78
Justification of choice of data or description of measurement methods and procedures applied	Derived from forest inventory data, IDEAM. See VCS Annex.
Purpose of data	Calculation of baseline emission
Comments	Emissions factors for estimating GHG emissions from deforestation.

Data / Parameter	Rj
Data unit	Relation factor
Description	Root shoot ratio
Source of data	IPCC/Literature value
Value applied	0.24
Justification of choice of data or description of measurement methods and procedures applied	Default value of 0.24 from IPCC Guidelines for National Greenhouse Gas Inventories. 2006. Table 4.3/Mokany 2006
Purpose of data	Calculation of baseline emission

Comments	Belowground biomass estimation
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Data / Parameter	Cbbcl
Data unit	t CO2e ha-1
Description	Average carbon stock per hectare in the below-ground biomass carbon pool of LU/LC class cl
Source of data	ER sheet
Value applied	forest class: bh-M=60.18 bh-MB=122.05 bh-PM=47.19 bmh-M=51.91 bmh-MB=107.58 bmh-PM and bmh-PMt=75.65 bms-T=40.62 bp-M=44.01 bp-MB=43.55 bp-PM=95.75 bs-T=39.79
Justification of choice of data or description of measurement methods and procedures applied	Calculated by applying the default value of 0.24 from IPCC Guidelines for National Greenhouse Gas Inventories. 2006.
Purpose of data	Calculation of baseline emission
Comments	Emissions factors for estimating GHG emissions from deforestation.

Data / Parameter	Ctot(icl)
Data unit	t CO2e ha-1
Description	Average carbon stock per hectare in the below-ground biomass carbon pool of LU/LC class cl
Source of data	ER sheet
Value applied	forest class: bh-M=310.92 bh-MB=630.61 bh-PM=243.82 bmh-M=268.19 bmh-MB=555.82

	bmh-PM y bmh-PMt=390.85 bms-T=209.85 bp-M=227.37 bp-MB=225.02 bp-PM=494.70 bs-T=205.57
Justification of choice of data or description of measurement methods and procedures applied	Derived from various forest inventory data. See ER sheet
Purpose of data	Calculation of baseline emission
Comments	Emissions factors for estimating GHG emissions from deforestation.

Data / Parameter	Ctoticl,t
Data unit	t CO2e ha-1
Description	Average carbon stock of all accounted carbon pools in forest class icl at time t
Source of data	Calculated, see Table Significancia, Annex VM0015
Value applied	Deemed de-minimus
Justification of choice of data or description of measurement methods and procedures applied	Significance analysis.
Purpose of data	Calculation of baseline emission
Comments	N.A

Data / Parameter	Cabfcl
Data unit	t CO2e ha-1
Description	Average carbon stock per hectare in the above-ground biomass carbon pool of final post-deforestation class fcl
Source of data	Calculated, see table CarbonPostdef, Annex GEI DB
Value applied	Grassland: 17.95 Heterogeneous farmland: 23.74 Crops: 21.78

Justification of choice of data or description of measurement methods and procedures applied	Calculated according to requirements of VM0015 v1.1.
Purpose of data	Calculation of baseline emission
Comments	Calculate GHG emissions from deforestation

Data / Parameter	Cp
Data unit	t CO2e ha-1
Description	Average carbon stock per hectare in the carbon pool p
Source of data	Table 17
Value applied	58.045
Justification of choice of data or description of measurement methods and procedures applied	Requirements of the VM0015 sec. 6.1.2.
Purpose of data	Calculation of baseline emission
Comments	Baseline GHG emissions estimates

Data / Parameter	Ctotfcl, t
Data unit	t CO2e ha-1
Description	Average carbon stock of all accounted carbon pools in non-forest class fcl at time t;
Source of data	N.A
Value applied	N.A
Justification of choice of data or description of measurement methods and procedures applied	Leakage management activities do not decrease carbon stocks.
Purpose of data	Calculation of baseline emission
Comments	N.A

Data / Parameter	$\Delta C_{abABSLKK}$
Data unit	t CO2e

Description	Cumulative baseline carbon stock changes for the above-ground biomass pool in the leakage belt
Source of data	Table 21.c.1, Annex VM0015
Value applied	33,927,167
Justification of choice of data or description of measurement methods and procedures applied	GHG accounting in the leakage belt.
Purpose of data	Calculation of leakage
Comments	N.A

Data / Parameter	$\Delta C_{bb}ABSLKK$
Data unit	t CO ₂ e
Description	Cumulative baseline carbon stock changes for the below-ground biomass pool in the leakage belt
Source of data	Table 21.c.2 Annex VM0015
Value applied	See Table 21.c.2 Annex VM0015 4,521,055
Justification of choice of data or description of measurement methods and procedures applied	GHG accounting in the leakage belt.
Purpose of data	Calculation of leakage
Comments	N.A

Data / Parameter	$\Delta C_{ab}BSLPA$
Data unit	t CO ₂ e
Description	Cumulative baseline carbon stock changes for the above-ground biomass pool in the project area
Source of data	Table 21.b.1 Annex VM0015
Value applied	See Table 21.b.1 Annex VM0015 4,556,726

Justification of choice of data or description of measurement methods and procedures applied	GHG accounting in the project area.
Purpose of data	Calculation of baseline emissions
Comments	N.A

Data / Parameter	$\Delta C_{bb}ABSLPA$
Data unit	t CO ₂ e
Description	Cumulative baseline carbon stock changes for the below-ground biomass pool in the project area
Source of data	Table 21.b.1 Annex VM0015
Value applied	See Table 21.b.1 Annex VM0015 431,903
Justification of choice of data or description of measurement methods and procedures applied	GHG accounting in the project area.
Purpose of data	Calculation of baseline emissions
Comments	N.A

Data / Parameter	$\Delta CADLK$
Data unit	t CO ₂ e
Description	Cumulative total decrease in carbon stocks due to displaced deforestation
Source of data	Table 36 Annex VM0015
Value applied	0
Justification of choice of data or description of measurement methods and procedures applied	GHG accounting from displaced leakage
Purpose of data	Calculation of leakage
Comments	N.A

Data / Parameter	ΔCBSLPA
Data unit	t CO ₂ -e
Description	Total baseline carbon stock changes in the project area
Source of data	Table 36, Annex VM0015
Value applied	See Table 36, Annex VM0015 7,197,895
Justification of choice of data or description of measurement methods and procedures applied	GHG accounting in the project area
Purpose of data	Calculation of leakage
Comments	N.A

Data / Parameter	ΔCPSPA
Data unit	t CO ₂ -e
Description	Cumulative project carbon stock change within the project area at year t
Source of data	See Tables 27 and 36, Annex VM0015
Value applied	Tables 27 and 36, Annex VM0015 359,895
Justification of choice of data or description of measurement methods and procedures applied	Calculation of net GHG emissions reductions
Purpose of data	Calculation of baseline emissions
Comments	N.A

Data / Parameter	ΔCUDdPA
Data unit	t CO ₂ -e
Description	Cumulative actual carbon stock change due to unavoided unplanned deforestation at year t in the project area
Source of data	Table 27, Annex VM0015.
Value applied	359,895

Justification of choice of data or description of measurement methods and procedures applied	Assumption of project effectiveness
Purpose of data	Calculation of project emissions
Comments	N.A

Data / Parameter	ΔREDDt				
Data unit	t CO2-e				
Description	Net anthropogenic greenhouse gas emission reduction attributable to the AUD project activity at year t				
Source of data	See Table 36, Annex VM0015				
Value applied	Table 36, Annex VM0015 <table><tr><td>11,92,859</td></tr><tr><td>16,02,949</td></tr><tr><td>18,59,015</td></tr><tr><td>18,11,653</td></tr></table>	11,92,859	16,02,949	18,59,015	18,11,653
11,92,859					
16,02,949					
18,59,015					
18,11,653					
Justification of choice of data or description of measurement methods and procedures applied	The cumulative result of applying the VM0015 methodology				
Purpose of data	Calculation of project emissions				
Comments	Final GHG calculations				

Data / Parameter	DLF
Data unit	%
Description	Displacement leakage factor
Source of data	Table 34, Annex VM0015
Value applied	5
Justification of choice of data or description of measurement methods and procedures applied	ex-ante leakage
Purpose of data	Calculation of leakage

Comments	N.A
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Data / Parameter	EI
Data unit	%
Description	ex-ante estimated Effectiveness Index
Source of data	Estimate generated by the project
Value applied	0.40
Justification of choice of data or description of measurement methods and procedures applied	Estimate generated by the project
Purpose of data	Calculation of project emissions
Comments	N.A

Data / Parameter	ELK
Data unit	t CO2-e
Description	Cumulative sum of ex-ante estimated leakage emissions at year t
Source of data	Table 36 Annex VM0015
Value applied	0
Justification of choice of data or description of measurement methods and procedures applied	The cumulative result of applying the VM0015 methodology
Purpose of data	Calculation of leakage
Comments	N.A

Data / Parameter	RFt
Data unit	%
Description	Risk factor used to calculate VCS buffer credits
Source of data	VCS Non Permanence Risk analysis
Value applied	11.75

Justification of choice of data or description of measurement methods and procedures applied	see VCS Non-Permanence Risk Analysis
Purpose of data	Calculation of project emissions
Comments	N.A

Data / Parameter	VBCt								
Data unit	t CO2-e								
Description	Number of Buffer Credits deposited in the VCS Buffer at time t;								
Source of data	See Table 36, Annex VM0015								
Value applied	Table 36, Annex VM0015 <table border="1"> <tr> <td>2016</td><td>134,304</td></tr> <tr> <td>2017</td><td>180,583</td></tr> <tr> <td>2018</td><td>209,554</td></tr> <tr> <td>2019</td><td>204,382</td></tr> </table> Total = 801,786	2016	134,304	2017	180,583	2018	209,554	2019	204,382
2016	134,304								
2017	180,583								
2018	209,554								
2019	204,382								
Justification of choice of data or description of measurement methods and procedures applied	Calculated								
Purpose of data	Buffer calculation								
Comments	N.A								

3.1.2 Data and Parameters Monitored

Data / Parameter	ABSLKt
Data unit	Ha
Description	Annual area of deforestation within the leakage belt at year t
Source of data	Satellite images

Description of measurement methods and procedures to be applied	Friends of the Riverine will be in charged for the climate monitoring according to the methodology VM0015 v1.1. Table 9c, 11c, 13c of Annex VM0015.								
Frequency of monitoring/recording	At each verification period								
Value applied	<table> <tr> <td>2016</td><td>17,092</td></tr> <tr> <td>2017</td><td>16,540</td></tr> <tr> <td>2018</td><td>16,216</td></tr> <tr> <td>2019</td><td>16,330</td></tr> </table>	2016	17,092	2017	16,540	2018	16,216	2019	16,330
2016	17,092								
2017	16,540								
2018	16,216								
2019	16,330								
Monitoring equipment	GIS software								
QA/QC procedures to be applied	Quality control/quality assurance (QA/QC) procedures will be performed by PP								
Purpose of data	Activity data for calculating GHG emissions. Calculated according to requirements of VM0015 v1.1.								
Calculation method	Calculation of leakage								
Comments	N.A								

Data / Parameter	ABSLPA
Data unit	Ha
Description	Cumulative area of deforestation within the project area at year t
Source of data	Satellite images
Description of measurement methods and procedures to be applied	Friends of the Riverine will be in charge of the climate monitoring according to the methodology VM0015 v1.1. Table 9b, 11b, 13b of Annex VM0015.
Frequency of monitoring/recording	At each verification period
Value applied	9,725

Monitoring equipment	GIS software
QA/QC procedures to be applied	Quality control/quality assurance (QA/QC) procedures will be performed by PP
Purpose of data	Activity data for calculating GHG emissions. Calculated according to requirements of VM0015 v1.1.
Calculation method	Calculation of project emissions
Comments	N.A

Data / Parameter	ABSLPA _t									
Data unit	Ha									
Description	Annual area of deforestation in the project area at year t									
Source of data	Satellite images									
Description of measurement methods and procedures to be applied	Friends of the Riverine will be in charged for the climate monitoring according to the methodology VM0015 v1.1. Table 9b, 11b, 13b of Annex VM0015.									
Frequency of monitoring/recording	At each verification period									
Value applied	<table><tr><td>2016</td><td>1,695.9</td></tr><tr><td>2017</td><td>2,248.2</td></tr><tr><td>2018</td><td>2,571.6</td></tr><tr><td>2019</td><td>2,457.8</td></tr></table>		2016	1,695.9	2017	2,248.2	2018	2,571.6	2019	2,457.8
2016	1,695.9									
2017	2,248.2									
2018	2,571.6									
2019	2,457.8									
Monitoring equipment	GIS software									
QA/QC procedures to be applied	Quality control/quality assurance (QA/QC) procedures will be performed by Friends of the Riverine									
Purpose of data	Activity data for calculating GHG emissions. Calculated according to requirements of VM0015 v1.1.									
Calculation method	Calculation of project emissions									

Comments	N.A
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Data / Parameter	ΔCPSPAt				
Data unit	t CO2-e				
Description	Annual project carbon stock change within the project area at year t				
Source of data	Satellite images and carbon stocks defined in 4.1				
Description of measurement methods and procedures to be applied	Friends of the Riverine will be in charge of the climate monitoring according to the methodology VM0015 v1.1. Table 27 and Table 36, Annex VM0015.				
Frequency of monitoring/recording	At each verification period				
Value applied	<table><tr><td>(60,290)</td></tr><tr><td>(81,062)</td></tr><tr><td>(94,064)</td></tr><tr><td>(91,738)</td></tr></table>	(60,290)	(81,062)	(94,064)	(91,738)
(60,290)					
(81,062)					
(94,064)					
(91,738)					
Monitoring equipment	N.A				
QA/QC procedures to be applied	Quality control/quality assurance (QA/QC) procedures will be performed by Friends of the Riverine				
Purpose of data	Calculation of project emissions				
Calculation method	Activity data for calculating GHG emissions reductions.				
Comments	N.A				

Data / Parameter	Δ REDD
Data unit	t CO ₂ -e
Description	Cumulative net anthropogenic greenhouse gas emission reduction attributable to the AUD project activity
Source of data	Methodology VM0015 v1.1. Table 36, Annex VM0015
Description of measurement methods and procedures to be applied	According to the methodology VM0015 v1.1. Table 36, Annex VM0015.
Frequency of monitoring/recording	At each verification period
Value applied	71,09,545
Monitoring equipment	N.A
QA/QC procedures to be applied	Friends of the Riverine will assign a QA/QC coordinator
Purpose of data	Calculation of project emissions
Calculation method	Final GHG calculations
Comments	N.A

Data / Parameter	Δ REDD _t
Data unit	t CO ₂ -e
Description	Net anthropogenic greenhouse gas emission reduction attributable to the AUD project activity at year t
Source of data	Methodology VM0015 v1.1. Table 36, Annex VM0015
Description of measurement methods and procedures to be applied	According to the methodology VM0015 v1.1. Table 36, Annex VM0015
Frequency of monitoring/recording	At each verification period

Value applied	
	831,853
	1,124,361
	1,311,585
	1,288,455
Monitoring equipment	N.A
QA/QC procedures to be applied	Friends of the Riverine will assign a QA/QC coordinator
Purpose of data	Calculation of project emissions
Calculation method	Final GHG calculations
Comments	N.A

Data / Parameter	RFt
Data unit	%
Description	Risk factor used to calculate VCS buffer credits
Source of data	VCS Non Permanence Risk analysis
Description of measurement methods and procedures to be applied	AFOLU Non permanence Risk Tool v.3.2
Frequency of monitoring/recording	At each verification period
Value applied	11.75
Monitoring equipment	N.A
QA/QC procedures to be applied	N.A
Purpose of data	Calculation of project emissions
Calculation method	N.A

Comments	Buffer calculation
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Data / Parameter	VBCt				
Data unit	t CO2-e				
Description	Number of Buffer Credits deposited in the VCS Buffer at time t;				
Source of data	Methodology VM0015 v1.1. Table 36, Annex VM0015				
Description of measurement methods and procedures to be applied	According to the methodology VM0015 v1.1. Table 36, Annex VM0015				
Frequency of monitoring/recording	At each verification period				
Value applied	<table><tr><td>134,304</td></tr><tr><td>180,583</td></tr><tr><td>209,554</td></tr><tr><td>204,382</td></tr></table>	134,304	180,583	209,554	204,382
134,304					
180,583					
209,554					
204,382					
Monitoring equipment	N.A				
QA/QC procedures to be applied	N.A				
Purpose of data	Calculation of project emissions				
Calculation method	N.A				
Comments	Buffer calculation				

Data / Parameter	VCUt
Data unit	t CO2-e
Description	Number of Verified Carbon Units (VCUs) to be made available for trade at time t
Source of data	Methodology VM0015 v1.1. Table 36, Annex VM0015

Description of measurement methods and procedures to be applied	According to the methodology VM0015 v1.1. Table 36, Annex VM0015				
Frequency of monitoring/recording	At each verification period				
Value applied	<table><tr><td>738,157</td></tr><tr><td>997,614</td></tr><tr><td>1,163,611</td></tr><tr><td>1,142,927</td></tr></table>	738,157	997,614	1,163,611	1,142,927
738,157					
997,614					
1,163,611					
1,142,927					
Monitoring equipment	N.A				
QA/QC procedures to be applied	N.A				
Purpose of data	Calculation of project emissions				
Calculation method	N.A				
Comments	Buffer calculation				

3.1.3 Monitoring Plan

Quality control and quality assurance procedures:

As monitoring was conducted mainly by using remote sensing imagery in a GIS environment, quality management procedures are related with GIS data quality. When working with geographical information, positional accuracy is often synonymous with data quality. Data compilation from several different sources often requires very good data quality.

Quality procedures must be employed to ensure that data are properly collected, handled, processed, used, and maintained throughout the data lifecycle while performing monitoring activities.

For Satellite Images, GIS and GPS data:

Quality Management Procedures before data collection:

- Monitoring was done by trained project personnel, understanding the importance of accurate data collecting.
- Format: Satellite Images were collected in TIFF raster digital format. Vector files were collected in shape file GIS format. GPS was also collected in digital format.

- Spatial Reference: Data was always use the Projected UTM System, South 22 Zone, Datum WGS84.
- Imagery: Landsat LCDM (Landsat 8) is the type of image selected for permanent monitoring.
- Source: Imagery were downloaded from the US Geologic Survey website (currently <http://glovis.usgs.gov/>).
- Responsibility for quality assurance was assigned to a person with adequate background suitable to the task.
- References on GPS and satellite images calibration are described in this document in the section "Equipment calibration".

Metadata: Metadata was created for the data to be collected.

GPS accuracy: To obtain accuracies such as those reported, the GPS Receiver must be located in an area that is free from overhead obstructions and reflective surfaces and have a good field of view to the horizon (for example, they do not work very well under a heavy forest canopy). Data coordinates were collected in meters using the spatial reference units noted above.

The staff member for monitoring evaluates the data gathering team to identify errors in field techniques, verify measurements processes and correct any identified problems before measurements are carried out.

Quality Management Procedures during data entry:

- Protocols and Guides: Data in digital formats must be treated according to protocols on data management. Methodology to process and classify satellite images must be carried out according to the annexed section on "Classification and Accuracy Assessment".
- Design an efficient storage system for the data: Data was stored in a computerized
 - database for use with specialized software (GPS, GIS, Remote sensing) for processing it. One person was in charge of managing this dataset.
- All modifications to the dataset shall be documented.

Quality Management Procedures after data entry:

- Regular backup of data helps ensure consistent quality levels.
- Data integrity: Data was not altered or destroyed in an unauthorized manner, or accidentally or maliciously modified.
- Data integration: Geographical datasets are difficult to integrate when there are inconsistencies between them, which may involve both the spatial and attributes characteristics of the data.
- Given the timeframe of the project, and the pace of production of updated versions of software and new hardware for storing data, electronic copies of data and reports were periodically updated or converted to a format that can be assessed by any future software applications.

All activities included in the Monitoring Plan are documented, as guides or protocols to be used permanently. It is required a data quality testing in each phase of data capture, including validation of GPS coordinates measured in the field (at a level community scale), choice of relevant data from a relevant source, positional accuracy of satellite images and GIS data, using automated GIS software tools and accuracy assessment of classified images.

Positional accuracy in satellite images

It is very important to ensure that remote sensing images in a time series overlay properly to each other and to other GIS maps used in the analysis. To geometrically correct the acquired images, it is necessary to have at least one orthorectified image to use as a reference image, and to undertake a process called geo-referencing.

Images were geometrically corrected to the projection UTM Zone 22 South datum WGS 1984 using an orthorectified image by collecting manually selected control points in both images. Among the features to be identified as control point locations we have road or stream intersections, the mouth of a stream, the corner of an established plot, or roads corners. The second-order polynomial transformation and the nearest neighbor resampling technique could be used to correct images. RMS errors of less than one pixel obtained from the geometric correction process was acceptable.

Accuracy of the baseline deforestation

Accuracy assessment follows the guidelines of the document „Methodology Procedures Used in Processing, Classification, and Accuracy Assessment of Remotely Sensed Data“ included the excel sheet submitted to VVB.

GHG emissions calculations

The tables to calculate GHG emission benefits from the project have been reviewed during the verification process and errors have been corrected. A cell by cell review was performed to follow the formulas and values to make sure all were correct.

The Operational and Management Structure and responsibilities for the Project was divided into three basic departments: General Management, Technical Management and Operators (Figure 4).

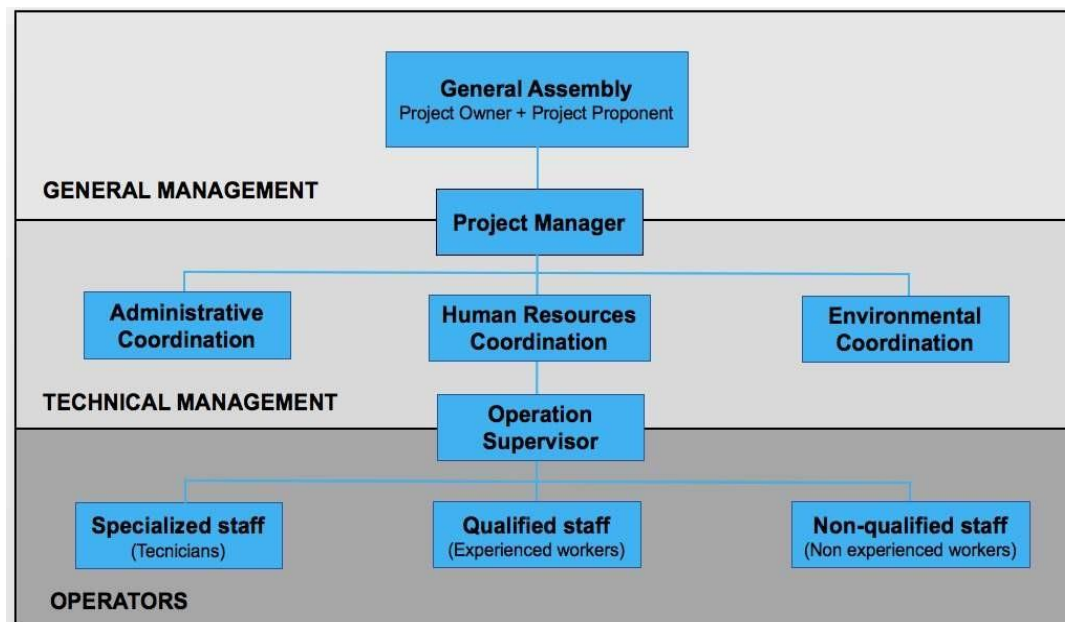


Figure 4: Operational and Management Structure.

Data Collecting Process

Random sampling method was chosen for ecological data collection, procedure is as explained: select one point randomly as your starting point. Then select the next point using a random number generator, and so on and so forth, until the required number of plots is reached. Plot size and location varies between 100 to 1000 sq.mts. Team recorded the plots by taking GPS coordinates and marked the boundary as well. Plots can be square or rectangular or circular depending on the geography.

Field sampling:

Sample size or number of plots to be established was based on the following calculation:

$$n = \frac{t^2 s^2}{E^2}$$

Where:

n = samples to estimate $\pm E$

t = t-value from student t-test table

s = standard deviation from prior/ trial run works

E = standard error of the mean

The project accounting area was divided into eight main clusters. Within each cluster sample plots were randomly established. All plots had their GPS coordinates recorded. The sampling strategy used was cluster sampling in which 10 to 32 plots were randomly clustered together in one area. Within the cluster the plots were spaced at a minimum distance of 1 to 2 km.

A total of 120 sample plots were the ideal number of plots to generate a representative data set based on the calculation. However, in the field one-hundred-forty sample plots were established of which seventy-five percent (75%) of the sample plots were converted to permanent sample plots (PSP).

Entire field process was carried out as mentioned in the carbon inventory report of the project.

Once the personal was contracted for monitoring the Project activities, the data collecting process started. For this component of the Project, permanent plots have been the base of data collecting. Data have been collected by the field crews that were composed by at least four members. The crew was composed by four forestry experts (qualified staff), one of them being the crew leader, and one additional person (qualified or not) that is familiarized with the local site, routes, strata, etc. The responsibilities for each crew member are summarized in Table 1.

Table 1: Crew members' responsibilities.

Crew member	Responsibilities
Forestry expert (Crew leader)	Organizing all the phases of the fieldwork, from the preparation to the data collection. This includes: preparing the fieldwork by carrying out bibliographic research, preparing field forms and maps, plan the work for the crew, administer the location of plots, ensure that field forms are properly filled in and that collected data is reliable, organize meetings after fieldwork in order to sum up daily activities, and organize working safety plan. When needed, crew leader will be in charge of training the crew for guarantying the accuracy of the data collection. Additionally, he had to contact and maintain good relationship with the survey manager and overview the progress achieved in the fieldwork.
Forestry expert (assistant of the crew leader)	Help the crew leader to carry out his/her task, take necessary measurements and observations, make sure that the equipment of the crew is always complete and operational, and supervise and guide workers.
Forestry expert	Measure DBH, height, litter, deadwood
Local person	Help to measure distances, facilitate access and visibility to technicians and inform about access to the strata.

TASK 1: Monitoring changes in carbon stocks and GHG emissions for periodic verification

- 1.1 Monitoring actual changes in carbon stocks and GHG emissions in the project area.
- 1.2 Monitoring leakage.
- 1.3 Ex-post calculation of GHG emission reductions.
- 1.4 Monitoring the impacts of natural disturbances and other catastrophic events.
- 1.1 Monitoring actual changes in carbon stocks and GHG emissions in the project area.
- 1.1.1 Monitor project implementation

This task of developing the monitoring report was the responsibility of the landowners. The landowner was in charge of sending patrol to the project area to identify loggers and squatters. The monitoring activities are done by STA solutions located in Belem.

Monitoring was conducted mainly by using remote sensing imagery and forest inventory made for the project. Remotely sensed data have been widely used as a cost-effective tool in the mapping and monitoring of large areas (e.g., Danaher et al. 1998; Gould 2000; Mayaux et al. 2000; Freeman et al. 2002). Satellite optical or radar images could be used depending on the availability of the scenes, cloud cover and related acquisition and processing costs. Satellite images need be calibrated (pre-processing procedures) before performing the principal analysis. These preprocessing operations could include (1) radiometric preprocessing to adjust digital values for effects of a hazy atmosphere and/or (2) geometric preprocessing to bring an image into registration with a map or another image (Campbell and Wynne, 2011).

At the same time, monitoring patrols composed by local people that patrol the Project Area to dissuade any invasion or illegal logging attempt. Fieldwork was done using a ground-based GPS approach to register the geographical position of any activity that need to be reported.

The Project Area was divided in patrols to better manage the extent of the Project Area and LMA and assure an effective on the ground monitoring. Patrol leaders oversee the activities, compile, and analyze the results from monitoring patrols³ (Figure 2). Patrol leaders report to the office in Belem. These two offices were

responsible of informing local authorities about illegal activities happening in the Project Area and to follow up the enforcement of the required measures to remove invaders (Figure 3).

EQUIPMENT CALIBRATION

GPS

Besides any operational considerations when using a GPS receiver in the field, this equipment need not be calibrated for this monitoring. Average positional accuracy of most GPS receivers (15 meters) is enough to register monitoring activities in the field (i.e., location of loggers or squatters in the Project Area). GPS coordinates were used 'for information only' and its accuracy is considered appropriate to clearly show in maps where any feature or activity in the Project Area is located.

Satellite Images

Monitoring was conducted mainly by using remote sensing imagery. Before processing the image data to produce land cover maps, satellite imagery need be calibrated, this is, some preprocessing procedures must be applied to the imagery. These procedures enhance the quality of the image data by reducing or eliminating various radiometric and geometric errors caused by internal and external conditions.

a. Geometric correction procedures address errors in the relative position of pixels due to factors such as variation in altitude, attitude and velocity of the sensor platform, Earth curvature, panoramic distortion, relief displacement and non-linearities in the sweep of a sensor (Lillesand and Kiefer 1994). It is very important to ensure that images in a time series overlay properly to each other and to other GIS maps used in the analysis. To geometrically correct the images, it is necessary to have a spread of Ground Control Points (GCPs) located either on a 1:100,000 official topography or another rectified image, to undertake a process called geo-referencing. For all resampling operations, we use the Nearest Neighbor algorithm to maintain radiometric integrity of the image.

b. Radiometric correction procedures account for errors that affect the brightness value of pixels due to both a sensor system detector error and an environmental attenuation error (e.g. changes in scene illumination, atmospheric conditions and viewing geometry [Lillesand and Kiefer 1994]). One of these procedures involved conversion of the measured multi-spectral brightness values to top of atmosphere reflectance units. This normalization procedure is crucial when creating multi-temporal and/or multi-spatial mosaics as it largely removes variations between these images due to sensor differences, Earth-sun distance and solar zenith angle (caused by different scene dates, overpass time and latitude differences) (Bruce and Hilbert, 2004).

The activities of the Project and their monitoring can be grouped as follows:

1. Forest monitoring:

Monitoring of forest cover was done mainly by remote sensing imagery and as per the forest inventory report made for the project. The choice of imagery depends on the availability of scenes, cloud cover, and related acquisition and processing costs. Remote sensing imagery could be either satellite (i.e Landsat) or radar (i.e Alos Palsar) or a combination of both.

LULC-change analysis using remote sensing imagery could be conducted annually or by periods depending on the availability of scenes. Therefore, verification event could occur annually or by periods.

The fixed degradation frontier map was updated at each verification event to account for areas that may have changed their carbon stocks due to not avoided illegal logging activities. The fixed degradation frontier was assessed based on the availability of scenes, cloud cover, and related acquisition and processing costs. Remote sensing imagery could be either satellite (i.e., Landsat) or radar (i.e., Alos Palsar) or a combination of both.

The landowners did limited patrols from 2012 to 2016 due to lack of money. These patrols would generate a hand-written report in Portuguese and then a law firm hired by the landowner would clean the reports and translate these to English. During this period no GPS points were taken to identify the exact location of deforestation activities. All original hand-written reports and the official translations are available in digital formats.

Prior to July 2016, patrolling activities were scaled-up by hiring local people as forest monitoring patrols. They were responsible for each monitoring patrol who generate monthly reports unless illegal activities are spotted, in which case a report was submitted immediately as described in the next section.

Of particular importance is the implementation of the surveillance system that allows continuous monitoring of the Project Area to prevent the entry of squatters and illegal loggers. The objective of forest patrolling is to make evident the presence of the landowner and dissuade pioneer agents (i.e., loggers, squatters) from encroaching the Project Area. This activity functions as a complement of remote sensing-based monitoring but does not replace it. Furthermore, the generation of carbon credits for avoided emissions does not rely on the results of patrolling,

The Project Area was divided in patrols (based on river affluents) to facilitate monitoring such a large area. Each Patrol leader compiles the information from patrolling reports into adequate digital format⁵ all the information from the reports to keep track of the areas that are being patrolled each week and what are the findings of each patrol. Digital reports were sent to the office in Belem to be organized and stored.

The actual size of a patrol and the numbers of community members to be hired depends on the results from the census that were conducted after validation. The area of a patrol is reasonable to allow for an effective surveillance given available staff. If not enough local community members are willing to work as monitoring staff, Ribeirinhos REDD+ via the Syndicate has hired people from Portel, Para urban area.

Patrols did identify illegal activities (invasions and timber extraction) and reported them the patrol leader. When illegal activities are spotted, patrols geo-reference the finding and make a short description of what was found. Patrols approach squatters or loggers to let them know –in good terms- that this is private land, they cannot undertake such activities there and they must leave immediately (virtual squatters are handled in the Brazilian legal system).

With the information supplied by each patrol, patrol leaders complete a monitoring report that include at least the coordinates where the illegal activities are taking place, the date and a brief report of what was identified. Finally, each patrol leader did submit this information to the local police in Portel and to IBAMA in Portel and in Belem.

Monitoring reports are numbered, filed appropriately, and be scanned to have digital copies in an archive as backup.

Once every two months, patrol leaders perform random site visits to verify that monitoring patrols are covering the assigned area and that each patrol is wearing the adequate field equipment. Patrol leaders complete a report that was submitted to office in Belem. A monitoring patrol that does not wear adequate field equipment or does not cover the designed monitoring route, did receive a warning. If a patrol is a reoffender, the patrol leader has a meeting with the offender to attempt to correct the issues.

Maps, reports, and records are available to validators at each verification event.

2. Biodiversity monitoring:

Monitoring will follow the guidelines described in section “B3. Biodiversity Impact Monitoring” of the associate CCB PDD.

Biodiversity activities and their indicators will be:

1. Biodiversity protection is directly related with ecosystem health, which is in turn linked to forest cover. Positive impacts on biodiversity will be estimated indirectly through remote analysis of forest cover. Preserving forest cover through avoided deforestation and degradation will allow implying that net positive impacts on biodiversity are being generated.
2. Biodiversity spotting by local people. Local people participating in the biodiversity component of the project will be in charge of reporting animal spotting at the boundaries of the LMA and the PA. Spotting frequency and animal species identified will indirectly assess net positive impacts on ecosystem health.
3. Active biodiversity monitoring. Local people will be hired to monitor ants, bats and bryophytes. Results are a good indirect indicator of ecosystem health.

All information should be properly reported following the protocols developed by The Ribeirinho REDD+ Project after the fieldwork. Reports should provide geo-referenced information about biodiversity spotting and data as determined by the protocols. All data from the reports should be input into electronic format prior to the analysis. Maps, reports and records will be available to validators at each verification event.

3. Social Monitoring:

Social Monitoring will be undertaken by social monitoring squads. There will be a responsible for each monitoring squad who will generate monthly activities reports. Each squad will be in charge of specific villages and will use approved questionnaires to gather socio-economic data about the impacts of the activities of the Project. Questionnaires will also include a section for comments to include information that is not contained in the template.

Monthly reports will be submitted to the brigade leader who will input the information into electronic format to analyze it. The results from this analysis will be used to assess the impact of each activity and to identify villages that require particular attention.

The Project management teams in Protel, and Belem will held bimonthly meetings to assess the effectiveness of the activities in local villages. Based on the information supplied by the brigade leaders, the management teams will improve the proposed activities. Maps, reports and records will be available to validators at each verification event.

1.1.2 Monitoring change and land use within the project area.

This task will be the responsibility of the Ribeirinho REDD+. According to the categories presented in Table 35 of the VM0015 methodology (Table 2), the Project will implement MRV to identify and assess LULC-changes within the Project Area.

Table 2: Categories that require MRV (refer to Table 35 – VM0015 methodology)

ID	Type	Conditions under which monitoring is mandatory	Explanations	Applicability to the Project
I	Area of forest land converted to non-forest land	Mandatory in all AUD project activities		Applicable
II	Area of forest land undergoing carbon stock decrease	Mandatory only for AUD project activities having planned logging,	Change in carbon stock must be significant according to ex-	Does not apply because none of the Project's activities involve

		fuel-wood collection and charcoal production activities above the baseline	ante assessment, otherwise monitoring is not required	planned logging, fuel-wood collection and charcoal production
III	Area of forest land undergoing carbon stock increase	Mandatory only for AUD project activities wishing to claim carbon credits for carbon stock increase	Increase must be significant according to ex-ante assessment and can only be accounted on areas that will be deforested in the baseline case	Does not apply because the project will not claim carbon credits from carbon stock increase.

Monitoring of forest cover will be done mainly by remote sensing imagery. The choice of imagery will depend on the availability of scenes, cloud cover, and related acquisition and processing costs.

Remote sensing imagery could be either satellite (i.e Landsat) or radar (i.e., Alos Palsar) or a combination of both.

LULC-change analysis using remote sensing imagery could be conducted annually or by periods depending on the availability of scenes. Therefore, verification event could occur annually or by periods.

LULC-change analysis will be done for the the Reference Region, Project Area and Leakage Belt using the protocol described in detail in the excel sheet submitted to VVB. This imagery classification analysis will result in forest and non-forest classes that should be compared with the results from the deforestation model for the date of a specific verification event. The results from such analysis will be reported using the appropriate VCS" tables and formats.

1.1.3 Monitoring of carbon stock changes and non-CO2 emissions from forest fires

None of the cases presented in Section 1.1.3 of the VM0015 methodology apply to the Project (Table 3). So, the Project is not required to set sampling plots to measure carbon stocks in either the project area or leakage belt.

Table 3: Applicability criteria for monitoring non-CO2 gasses

	ID	Type	Applicability to the Project
	Within the Project Area		
Mandatory monitoring of the carbon stocks	I	Areas subject to significant carbon stock decrease in the project scenario according to ex-ante assessment	Does not apply
	II	Areas subject to unplanned and significant carbon stock decrease e.g., due to uncontrolled forest fires and other catastrophic events	Does not apply

	III	Area of forest land undergoing carbon stock increase	Does not apply
	Within Leakage Management Area		
	IV	Areas subject to planned and significant carbon stock decrease in the project scenario according to ex-ante assessment	Does not apply
Option monitoring of carbon stocks	Within the Project Area		
	V	Areas subject to carbon stock increase after planned harvest activities	Does not apply
	VI	Areas recovering after disturbances	Does not apply
	Within Leakage Management Areas		
	VII	Areas subject to carbon stock increase due to leakage prevention measures	Does not apply
	Within Leakage Belt		
	VIII	Areas undergoing significant changes in carbon stock	Does not apply

1.1.4 Monitoring of the impacts of natural disturbances and other catastrophic events

Catastrophic events are not expected in the Project Area or Leakage Belt. Nevertheless, if by any chance a catastrophic event presents during the Project's lifetime, such events will be evaluated and reported if significant. Monitoring will follow VM0015 Tables 20.f, 20.g, 21.f and 21.g to report reductions by catastrophic events.

1.1.5 Total ex post estimated actual net carbon stock changes and GHG emissions in the project area

All ex-post estimations in the Project Area will be summarized using the format of Table 24 of the VM0015 methodology.

3.1.4 Dissemination of Monitoring Plan and Results (CL4.2)

A monitoring plan for climate benefits along with the monitoring plan and results were communicated to the communities and other stakeholders via the website for the project where the Monitoring plan is listed.

In addition to this the technicians who travel to the land are required to carry all the most up-to-date documents and go through them with the communities at their request.

The head of each household was shown a hard copy.

3.2 Quantification of GHG Emission Reductions and Removals

3.2.1 Baseline Emissions

Total emissions in the baseline scenario for year 2020 are 285,836 tCO₂e as presented in Table 4.

Table 4: Total net baseline carbon stock change in baseline scenario in the Project area (table 21b of VCS VM0015)

Carbon stock changes in the above-ground biomass per initial forest class <i>icl</i>		Total carbon stock change in the above-ground biomass of the initial forest classes in the project area		Carbon stock changes in the above-ground biomass per post-deforestation zone <i>z</i>		Total carbon stock change in the above-ground biomass of post-deforestation zones in the project area		Total net carbon stock change in the above-ground biomass of the project area	
ID <i>icl</i> >	1	ΔCab BSLPA _{<i>icl,t</i>}	ΔCab BSLPA _{<i>icl</i>}	ID <i>iz</i> >	1	ΔCab BSLPA _{<i>z,t</i>}	ΔCab BSLPA _{<i>z</i>}	ΔCab BSLPA _{<i>t</i>}	ΔCab BSLPA
Name >		annual	cumulative	Name >		annual	cumulative	annual	cumulative
Project year <i>t</i>	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	Project year <i>t</i>	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e
0	0	0	0	0	0	0	0	0	0
1	(11,66,416)	(11,66,416)	(11,66,416)	1	-	-	-	(11,66,416)	(11,66,416)
2	(15,46,324)	(15,46,324)	(27,12,740)	2	3,281	3,281	3,281	(15,49,605)	(27,16,021)
3	(17,68,753)	(17,68,753)	(44,81,492)	3	7,631	7,631	10,912	(17,76,384)	(44,92,405)
4	(16,90,458)	(16,90,458)	(61,71,950)	4	12,607	12,607	23,519	(17,03,065)	(61,95,469)
5	(5,17,211)	(5,17,211)	(66,89,161)	5	17,362	17,362	40,881	(5,34,573)	(67,30,043)

Carbon stock changes in the below-ground biomass per initial forest class <i>icl</i>	Total carbon stock change in the below-ground biomass of the initial forest classes in the project area	Carbon stock changes in the below-ground biomass per post-deforestation zone <i>z</i>	Total carbon stock change in the below-ground biomass of post-deforestation zones in the project area	Total net carbon stock change in the below-ground biomass of the project area
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ID icl>	1	ΔCab BSLPA _{icl,t}	ΔCab BSLPA _{icl}	ID iz>	1	ΔCab BSLPA _{z,t}	ΔCab BSLPA _z	ΔCab BSLPA _t	ΔCab BSLPA
Name >		annual	cumulative	Name >		annual	cumulative	annual	cumulative
Project year t	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	Project year t	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e
0	0	0	0	0	0	0	0	0	0
1	(25,986)	(25,986)	(25,986)	1	-	-	-	(25,986)	(25,986)
2	(60,435)	(60,435)	(86,421)	2	3,281	3,281	3,281	(57,154)	(83,140)
3	(99,840)	(99,840)	(1,86,261)	3	7,631	7,631	10,912	(92,209)	(1,75,348)
4	(1,37,500)	(1,37,500)	(3,23,761)	4	12,607	12,607	23,519	(1,24,894)	(3,00,242)
5	(1,49,023)	(1,49,023)	(4,72,784)	5	17,362	17,362	40,881	(1,31,661)	(4,31,903)

Carbon stock changes in litter per initial forest class icl				Carbon stock changes in litter per post-deforestation zone z				Total net carbon stock change in litter of the project area	
ID icl>	1	ΔCab BSLPA _{icl,t}	ΔCab BSLPA _{icl}	ID iz>	1	ΔCab BSLPA _{z,t}	ΔCab BSLPA _z	ΔCab BSLPA _t	ΔCab BSLPA
Name >		annual	cumulative	Name >		annual	cumulative	annual	cumulative
Project year t	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	Project year t	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e
0	0	0	0	0	0	0	0	0	0
1	(13,397)	(13,397)	(13,397)	1	-	-	-	(13,397)	(13,397)
2	(17,761)	(17,761)	(31,158)	2	3,281	3,281	3,281	(14,480)	(27,877)
3	(20,316)	(20,316)	(51,474)	3	7,631	7,631	10,912	(12,685)	(40,562)
4	(19,416)	(19,416)	(70,890)	4	12,607	12,607	23,519	(6,810)	(47,371)
5	(5,941)	(5,941)	(76,831)	5	17,362	17,362	40,881	11,422	(35,950)

Project year t	Baseline carbon stock changes		Baseline GHG emissions	
	annual	Cumulative	annual	cumulative
	$\square CBSLPA_t$	$\square CBSLPA$	$EBBBSLPA_t$	$EBBBSLPA$

	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e
0	-	-	-	-
1	(12,05,799)	(12,05,799)	(49,842.41)	(49,842.41)
2	(16,21,239)	(28,27,038)	(66,076.35)	(1,15,918.76)
3	(18,81,277)	(47,08,315)	(75,581.01)	(1,91,499.77)
4	(18,34,768)	(65,43,083)	(72,235.37)	(2,63,735.14)
5	(5,74,730)	(7,117,813)	(22,101)	(285,836)

Total Baseline Emissions: 285,836 tCO₂e

3.2.2 Project Emissions

Emissions due to planned deforestation:

No emissions associated to planned deforestation were developed in the Project area from 1st April 2016 to 30th March 2021.

Emissions due to planned logging activities

No emissions associated to planned logging activities were developed in the Project area from 1st April 2017 to 31th March 2021.

Emissions due to planned fuel-wood and charcoal activities

Removals due to carbon stock increase of planned activities

No emissions associated to planned fuel-wood and charcoal activities were developed in the Project area. Removals due to carbon stock increase of planned activities

Carbon stock increase due to planned activities in areas that would be deforested in the baseline case is not considered.

Total ex post carbon stock decrease in the Project area

No carbon stock decrease associated to Project activities has occurred in the Project area from 1st April 2017 to 31st March 2021

Emissions due to unavoidable unplanned deforestation

A total of 2,245 hectares of unavoidable unplanned deforestation was observed within the Project area in this monitoring period.

Total of emissions related to unavoidable unplanned deforestation is 210,276 tCO₂e. Emissions due to forest fires and catastrophic events

No emissions associated to forest fires and catastrophic events have occurred in the Project area during present monitoring period.

Ex post estimated net carbon stock change in the Project area

Total ex post estimated carbon stock change in Project area under the Project scenario in this monitoring period is presented in Table 5.

Table 5. Ex post estimated net carbon stock change in the Project area under the Project scenario (Table 29 of VCS VM0015)

Project year <i>t</i>	Total ex ante carbon stock decrease due to planned activities		Total ex ante carbon stock increase due to planned activities		Total ex ante carbon stock decrease due to unavoided unplanned deforestation		Total ex ante carbon stock change		Total ex ante estimated actual non-CO ₂ emissions from fires in the project area	
	annual □CPAdPA _t	cumulative □CPAdPA	annual □CPAiPA _t	cumulative □CPAiPA	annual □CUDdPA _t	cumulative □CUDdPA	annual □CPSPA _t	cumulative □CPSPA	annual EBBPSPA _t	cumulative EBBPSPA
	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e
0	0	0	0	0	0	0	0	0	0	0
1	0	0	0	0	(60,290)	(60,290)	(60,290)	(60,290)	(2,492)	(2,492)
2	0	0	0	0	(81,062)	(1,41,352)	(81,062)	(1,41,352)	(3,304)	(5,796)
3	0	0	0	0	(94,064)	(2,35,416)	(94,064)	(2,35,416)	(3,779)	(9,575)
4	0	0	0	0	(91,738)	(3,27,154)	(91,738)	(3,27,154)	(3,612)	(13,187)
5	0	0	0	0	(32,741)	(359,895)	(32,741)	(359,895)	(1,105)	(14,292)

Non-CO₂ emissions from forest fires. Not subjected to monitoring and accounting.

Total Project Emissions = 14,292

3.2.3 Leakage

One source of leakage was monitored: leakage due to displacement activity.

Leakage due to displacement activity was monitored by mapping forest cover change in the leakage belt. As defined in the VCS Methodology VM0015, deforestation above the baseline in the leakage belt area will be considered activity displacement leakage. Activity data for the leakage belt area was determined using the same methods applied to deforestation mapping in the Project area.

Total ex post estimated leakage

Ex-post total net carbon stock changes in the leakage belt due to displacement of activity in this monitoring period are presented in Table 6. Leakage was calculated as the difference between the ex-post and ex-ante the assessment. As the result is <0, total ex post leakage was set to zero.

Total Leakage Emissions = 0 tCO₂e

3.2.4 Net GHG Emission Reductions and Removals

The net anthropogenic GHG emission reduction of the proposed AUD project activity is calculated as follows:

$$\Delta REDD_t = (\Delta CBSLPA_t + EBBBSLPA_t) - (\Delta CPSPA_t + EBBPSPA_t) - (\Delta CLK_t + ELK_t)$$

Where:

$\Delta REDD_t$ *Ex post* net anthropogenic greenhouse gas emission reduction attributable to the AUD project activity at year t ; tCO₂e

$\Delta CBSLPA_t$ Sum of baseline carbon stock changes in the project area at year t ; tCO₂e

$EBBBSLPA_t$ Sum of baseline emissions from biomass burning in the project area at year t ; tCO₂e

$\Delta CPSPA_t$ Sum of *ex post* estimated actual carbon stock changes in the project area at year t ; tCO₂e

Note: If $\Delta CPSPA_t$ represents a net increase in carbon stocks, a negative sign before the absolute value of $\Delta CPSPA_t$ shall be used. If $\Delta CPSPA_t$ represents a net decrease, the positive sign shall be used.

$EBBPSPA_t$ Sum of actual emissions from biomass burning in the project area at year t ; tCO₂e

ΔCLK_t Sum of *ex post* estimated leakage net carbon stock changes at year t ; tCO₂e

Note: If the cumulative sum of ΔCLK_t within a fixed baseline period is > 0 , ΔCLK_t shall be set to zero.

ELK_t Sum of *ex post* estimated leakage emissions at year t ; tCO₂e

t 1, 2, 3 ... T , a year of the proposed project crediting period; dimensionless.

The number of Verified Carbon Units (VCUs) to be generated through the proposed AUD project activity at year t is calculated as follows:

The net anthropogenic GHG emissions reductions were calculated following the equation 19, 20 and 21 of VCS VM0015 version 1.1. The risk factor used to calculate VCS buffer credits (VBC) is 11.75%, as calculated in Non-permanence Risk Report.

Table 6: Ex-ante estimated net anthropogenic GHG emission reductions (DREDDt) and Voluntary Carbon Units (VCUs)

Project year t	Baseline GHG Emissions	Project GHG Emissions	Leakage	Net Anthropogenic GHG emissions reductions	VCUs Tradable	Buffer Credits
1 st April 2017 – 31 st March 2018	34,446.33	1,722	-	831,853	738,157	93,695

1 st April 2018 – 31 st March 2019	45,665.69	2,283	-	1,124,361	997,614	126,747
1 st April 2019 – 31 st March 2020	52,234.41	2,612	-	1,311,585	1,163,611	147,974
1 st April 2020 – 31 st March 2021	49,922.22	2,496	-	1,288,455	1,142,927	145,528

Estimates of GHG credits eligible for issuance as VCUs were calculated in Table 6, where

Estimated GHG emission reduction credits = Baseline emissions, fixed for 10 years at validation minus, Project emissions minus, Leakage minus, Non-permanence Risk Buffer withholding (calculated as a percent of net change in carbon stocks prior to deduction of leakage)

Net Emissions Reduction 831,853 + 1,124,361 + 1,311,585 + 1,288,455

Net Emissions Reduction = 4,556,254 tCO₂e

$$VCUt = \Delta REDDt - VBCt$$

$$VBCt = (\Delta CBSLPAt - \Delta CPSPAt) \times RFt$$

Where:

$VCUt$ Number of Verified Carbon Units that can be traded at time t ; tCO₂e

Note: If $VCUt < 0$ no credits (VCUs) will be awarded to the proponents of the AUD project activity.
 $\Delta REDDt$ *Ex post* estimated net anthropogenic greenhouse gas emission reduction attributable to the AUD project activity at year t ; tCO₂e

$VBCt$ Number of Buffer Credits deposited in the VCS Buffer at time t ; tCO₂e

$\Delta CBSLPAt$ Sum of baseline carbon stock changes in the project area at year t ; tCO₂e

$\Delta CPSPAt$ Sum of *ex post* carbon stock changes in the project area at year t ; tCO₂e ha⁻¹

RFt Risk factor used to calculate VCS buffer credits; %

t 1, 2, 3 ... T , a year of the proposed project crediting period; dimensionless.

Ex-ante Verified Carbon Units = 738,157 + 997,614 + 1,163,611 + 1,142,927

Total ex-ante Verified Carbon Units = **4,042,310** tCO₂e

Table 7: Summary of estimated tradable net anthropogenic GHG emission reductions (DREDDt) and Voluntary Carbon Units (VCUs) after application of buffer

Project Year	VCU tradable	
	Vintage wise	Buffer applied
2017	738,157	11.75%
2017	997,614	11.75%
2019	1,163,611	11.75%
2020	1,142,927	11.75%

3.3 Optional Criterion: Climate Change Adaptation Benefits

Not applicable

3.3.1 Activities and/or processes implemented for Adaptation (GL1.3)

Not applicable

4 COMMUNITY

4.1 Net Positive Community Impacts

4.1.1 Community Impacts (CM2.1)

Community Group	- Riverine Community and traditional rural villagers - The entire community is all the riverine in the project area. - The community is broken into households, these households are better described as families. Each household may have up to 13 houses each being houses of the children of the original settler. The project has targeted to have 5 people trained in diversity.
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Impact	- Number of riverine people and traditional rural villagers participating in the monitoring - Improved livelihood - monitoring activities each month - One person in each river that is next to the project area, with a general awareness to all member in each community. - 5 People total, as 5 primary different rivers. - Team of technicians completing the survey work for each family are trained to biodiversity monitor and report back any unique events. - Over 50 people along 5 different river affluents. - 1,252 received land tenure document known as CAR
Type of Benefit/Cost/Risk	- Actual benefit. - Capacity building related to the monitoring and CCB management of the forest and biodiversity. - Very little cost, as the monitoring is done via cell phone camera when there is a unique event, such as a large carnivore enter the area in the vicinity of the houses. - Job Opportunities - Providing land ownership legal rights
Change in Well-being	- Improved livelihood – more aware of the fauna and flora. - Capacity building - Improved agricultural practices - Increase in safety and security

Community Group	Women of the community are seen as vulnerable group
Impact	- 500 women were trained on the efficient cook stoves. - 500 cook stoves have been provided to the riverine people. - Number of improved cooking stoves pilots implemented in local families - Capacity building related to efficient and improved cooking stoves - improvement in health
Type of Benefit/Cost/Risk	- Actual benefit - overall satisfaction and health of community
Change in Well-being	- Improvement in overall satisfaction and health of community

4.1.2 Negative Community Impact Mitigation (CM2.2)

There are no negative community impacts and hence there is no need for mitigation.

4.1.3 Net Positive Community Well-Being (CM2.3, GL1.4)

Net Positive community impacts for this monitoring period is summarized as:

- Secured land tenure
- Diversification of food through agroforestry practices thus an improvement in local nutrition
- More efficient technologies to produce farinha therefore less time is consumed in this activity.
- Generation of income from monitoring activities.
- Better understanding of the importance of protecting the forest and how forest conservation
- benefits their livelihoods.

As shown in Section 4.1.1, most of the activities initiated during the reporting period have multiple positive impacts on a large segment of the communities under the Riberibnho REDD+. The potential costs or negative impacts from implementing the proposed project activities are nil/minimal and are being mitigated for to some extent through the community projects (e.g., for poachers). Consequently, the net well-being impacts for the reporting period are overwhelmingly positive given the proportion of the population potentially impacted, the magnitude of the impacts and their long-term nature

4.1.4 Protection of High Conservation Values (CM2.4)

As shown under Section 4.1.1, the project activities undertaken during the reporting period were all geared towards reducing pressure on the forest by diversifying livelihoods away from direct natural resource exploitation and enhancing forest protection. No negative effects are observed because of the project activities.

No High Conservation Values (HCVs) were negatively affected as a result of the project. HCVs related to community identified by the Project are the better conditions of life and work of the communities linked to the Project itself.

4.2 Other Stakeholder Impacts

4.2.1 Mitigation of Negative Impacts on Other Stakeholders (CM3.2)

There are not expected negative offsite impacts thus no mitigation strategies are required.

4.2.2 Net Impacts on Other Stakeholders (CM3.3)

The Project activities have not resulted in net negative impacts on the well-being of other stakeholders because all the activities were conducted in a private area and in addition to that all the stakeholders were always consulted and the results of this consultation can be considered very positive.

4.3 Community Impact Monitoring

4.3.1 Community Monitoring Plan (CM4.1, CM4.2, GL1.4, GL2.2, GL2.3, GL2.5)

The project proponents have designed a Social Impacts Monitoring Plan in accordance to the results obtained in the rural participatory diagnosis developed in the project area and initially considering the indicators for the products of the proposed activities based on the identification of the necessities indicated by the population and the strategies foreseen to accomplish the project goals.

For presenting community impact monitoring's results few parameters were considered and the same has been presented in the table format. Table 8 below shows which takes the key impact factors for the community and other stakeholders. It was observed that the project area, before the start of the Project, there were just a farm worker and his family, that lived in the farm and who took care of the livestock. Nowadays the farm workers community living today in the farm are very happy to see the developments in and around the project area due to REDD project. In order to include the evaluation of the Project impacts with the farm workers' point of view, we gave them a questionnaire (Table 8) so they could complete it in

an absolutely anonymous way. For other stakeholders (commercial and institutional), instead, a verbal interview was conducted.

The following Table (Table 8) shows a non-comprehensive list of activities and indicators that were considered during monitoring. A full and detailed list is presented below:

Table 8: key impact factors for the community and other stakeholders

Category	Impact	Results and evaluation	Sampling method	Frequency and dates/Milestones
WORKERS COMMUNITY	Employment	The Project has created 60 permanent new jobs, 20 temporary jobs, employed 9 technicians, given employment up to 89 people depending on the month . The project intends to have employment at 120 staff by 2022. These workers perceive better salaries compared to those obtained in absence of the Project. Work condition are at good level: high safety, social security and unemployment insurance.	Questionnaire compiled by the workers. Data analysis prepared by Project Proponent.	December 2020: date of the questionnaire completion December 2020: data analysis by PP. Next evaluations: each verification period. Milestone: For the entire monitoring period.
	Training and education	Thanks to the numerous training sessions and workshops carried out during the monitoring period, workers have increased their technical skills and forestry knowledge.	Questionnaire compiled by the workers. Data analysis prepared by Project Proponent.	December 2020: date of the questionnaire completion December 2020: data analysis by PP. Next evaluations: each verification period. Milestone: For the entire monitoring period.
	Life conditions and health	Life conditions and health have been improved thanks to fair salaries, decent housing conditions, free transports to the city, internet connections, drinking water, internal road maintenance, horticulture and small-scale livestock breeding (chicken and pork's).	Questionnaire compiled by the workers. Data analysis prepared by Project Proponent.	December 2020: date of the questionnaire completion December 2020: data analysis by PP. Next evaluations: each verification period. Milestone: For the entire monitoring period.
INSTITUTIONAL STAKEHOLDERS	Knowledge	Sindicato of Rural Producers, Secretary of Education for the Municipality of Portel, Friends of the Riverine, as well as locals had the opportunity to learn about the Project, to keep update regarding the evolution of it, and to transfer important information to their network. Being the first carbon project of the region has aroused a strong interest from these institutions that recognize in it an excellent tool for sustainability.	Interviews made by PP.	December 2020: date of the questionnaire completion December 2019: data analysis by PP. Next evaluations: each verification period. Milestone: For the entire monitoring period.

	Opportunity	The strong interest regarding the Project has generated several requests of set-up carbon projects from small-medium forestry entrepreneurs.	Interviews made by PP.	December 2019: date of the questionnaire completion December 2019: data analysis by PP. Next evaluations: each verification period. Milestone: For the entire monitoring period.
COMMERCIAL STAKEHOLDERS	Knowledge	Friends of the Riverine had the opportunity to learn about the Project, to keep update regarding the evolution of the Project and to transfer important information to their network.	Interviews made by PP.	December 2019: date of the questionnaire. December 2019: data analysis by PP. Next evaluations: each verification period. Milestone: For the entire monitoring period.
	Business	Supply chain's stakeholder have had a positive economic impact having been able to benefit from various forms of work and revenues.	Interviews made by PP.	December 2020: date of the questionnaire completion December 2019: data analysis by PP. Next evaluations: each verification period. Milestone: For the entire monitoring period.

4.3.2 Monitoring Plan Dissemination (CM4.3)

The plan was provided to local access via the project webpage and they have this.

In addition to this the technicians who travel to the land are required to carry all the most up-to-date documents and go through them with the communities at their request. All the information was disseminated in their local language Portuguese.

The head of each household were shown a hard copy of the results and a discussion took place to make sure that this person understood it.

4.4 Optional Criterion: Exceptional Community Benefits

Not applicable.

5 BIODIVERSITY

5.1 Net Positive Biodiversity Impacts

5.1.1 Biodiversity Changes (B2.1)

The project focus exclusively on conservation measures within the project boundaries and its buffer, which makes negative offsite effects unlikely to happen. Besides, monitoring of flora and fauna this assures that

any minimal offsite negative effect was taken care of immediately. Also, as mentioned on G3.2, the Project's activities do not involve the introduction of non-native species and the engagement of local community contributes for the socio-environmental safeguarding activities.

It is also very unlikely that the Project's activities within its boundaries (implementation of agroforestry techniques, energy efficient cook stoves for farinha production, and tenure rights) have any offsite impact. Therefore, considering these activities and "with project" scenario, the effects of the project on biodiversity is positive.

Change in Biodiversity	Increase in biodiversity (FLORA)
Monitored Change	Analyzing the Project Area and the Project Zone with a quick and superficial examination it could be stated that the Rio REDD project represents a very good conservation initiative steps to conserve natural forests. Native amazon (composed of rain forest areas), it has been well preserved due to the presence of the Project, that nowadays constitute a real "mosaic of ecosystems" that clearly differentiate from the definition of deforestation. Also in the surrounding areas there is a real undergrowth, rich in botanical species, which thus enriches the Project Zone with biodiversity.
Justification of Change	Regarding the Amazon areas of the project, it can be stated that in recent years it has gone through a real conservation process. As described and tested later in the specific section on fauna, project has led to an increase in animal populations, to the fact that the native tree species were conserved, provide shelter to the animals and allow an easy and safe displacement for the same between the amazon areas. These areas have begun to benefit and regenerate after the beginning of the Project. In fact during the Project Period these areas have been geo-referenced, surrounded by fences (which limit the entry of people, machineries and cattle but do not prevent the displacement of wild animals) and it has been strictly prohibited entry to people, even to farm workers. Obviously hunting and other illegal activities were prohibited in these areas.

Change in Biodiversity	Total planned area for conservation
Monitored Change	At the time of current Monitoring Report, the planned conserved areas (Project Area) correspond to the initial planned area of 199,962ha (see 1.1.2.1 – Project Design of PDD). During the first 5 years and 0 months of Project, no loss of areas has ever occurred. Neither fires, nor floods, nor other climatic events have compromised the development of the trees that make up the forest.
Justification of Change	Periodic analysis of satellite imagery and GIS analysis and assumptions used to estimate or document the change. Greatly reduced illegal logging in the project area, and this is mostly

	allowed old logging roads to fill with vegetation. This elimination of illegal wood activities has helped the flora replenish The old logging roads from satellite footage from 2008 to the present have mostly filled in, on current 2020 satellite maps.
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Change in Biodiversity	Increase in number of are-limited species
Monitored Change	<u>Monitor area-limited species: species that require large patches to maintain viable populations, such as large carnivores. That indicates potential habitat losses and prey availability</u> Increase in number of specific category species noted by sightings during regular patrolling for this monitoring period Puma concolor/ onca-parda- 4 Panthera onca/ onca-pintada - 4 Manatee – 5
Justification of Change	Increase in number of specific category species as the project area is not disturbed by agents of deforestation. This is noted by sightings during regular patrolling

Change in Biodiversity	Increase in number of resource-limited species
Monitored Change	<u>Monitor resource-limited species: species requiring specific resources, such as frugivorous species, nectar species, snags etc. Bats can be great bio indicators as they have different feeding habits, such as insects, fruits, nectar/pollen, blood etc. They are also abundant through the region and its taxonomy has been well documented</u> Increase in number of specific category species noted by sightings during regular patrolling for this monitoring period: araçari-de-pescoço-vermelho - 1 João-teneném-castanho - 2 jacupiranga - 1 mãe-de-taoca – 1
Justification of Change	Increase in number of specific category species as the project area is not disturbed by agents of deforestation. This is noted by sightings during regular patrolling

Change in Biodiversity	Increase in number of special interest species
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Monitored Change	Increase in number of specific category species noted by sightings during regular patrolling for this monitoring period: <u>Monitor “special interest” species, critically endangered species, endangered species, and threatened species (IUCN, IBAMA)</u> Anta - 10 Queixada - 1 Graxaim - 4 macaco-da-noite - 1 Puma concolor/ onca-parda- 4 Panthera onca/ onca-pintada - 4
Justification of Change	Increase in number of specific category species as the project area is not disturbed by agents of deforestation. This is noted by sightings during regular patrolling

5.1.2 Mitigation Actions (B2.3)

The project during the monitoring period has sent teams of people, every year in two forms: Security and Monitoring, and the other team is training / implementing social projects (land tenure/cook stoves) and conservation education. The biodiversity monitoring is implemented as part of the regular patrolling conducted. The patrol guards note the sightings in the log book, which is reflected for this monitoring period. On an average 6 to 7 patrols have been conducted each year during this monitoring period. It is also informed that such patrols which have been on-going since project start date of 2009 onwards Security Patrols are coordinated by Silas, Camero and Sergio. Further mitigation actions such as creating the awareness on conservation issues to the community and steps to prevent and remove land grabbers and illegal logging activities have ensured that the habitat is not fragmented.

Samples of the patrol logs have been submitted to the VVB.

5.1.3 Net Positive Biodiversity Impacts (B2.2, GL1.4)

Net positive impacts on biodiversity have been demonstrated for this monitoring plan over time through periodic monitoring and reporting of biodiversity indicators as per the Biodiversity Monitoring Plan.

As already described in the PD, the Baseline Scenario was considered to be degraded land with damage forest cover, which was used for extensive cattle grazing, with very low meat production per hectare. The practice of breeding was totally irrational, “extractive” and not sustainable. Fire has often been used to stimulate pasture sprouting in the grazing areas and also in the Legal Reserve areas. Farm workers occasionally practiced hunting outside and inside the farm borders, including Legal Reserve areas. All these practices have brought to a drastic reduction in biodiversity over the years. In fact fauna, because of fire, the presence of livestock, hunting and the scarcity of food and shelter, was practically absent from the region. Also the flora was clearly compromised, because of degraded soil, use of fire and animal pressure that did not allow grazing to regenerate. The Project Scenario has led to a marked increase in biodiversity (flora and fauna) for all the arguments explained in the previous paragraphs.

5.1.4 High Conservation Values Protected (B2.4)

High Conservation Value	Caxiuna National Forest
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Qualifying Attribute	The Caxiuana National Forest is considered the oldest in the Amazon region and the second in Brazil. It is amongst the most known conservation units in north of Brazil, and it has the presence of many important researchers from Brazil and abroad. There have been no changes to the HCV area for this monitoring period as analyzed from GIS data. Further this serves as a biodiversity corridor resulting in the sightings of the endangered species in the project area.
Focal Area	On the northern border of the reference region there is a national conservation unit called National Forest Caxiuanã. It was created in 1961 and today it has an area of 322,694.34 hectares. The Conservation Units are types of conservation areas that were created to allow sustainable use of the forest and its natural resources

5.1.5 Invasive Species (B2.5)

None of the Project's activities introduce invasive species or genetically modified organisms. The Project's developer only approves agroforestry activities that use native species commonly known to occur in the Para region and are not in the Global Invasive Species Database before approving the utilization of particular species. Hence no invasive species were introduced for this monitoring period.

5.1.6 Impacts of Non-native Species (B2.6)

No non-native species was used in the Project Accounting Area.

5.1.7 GMO Exclusion (B2.7)

No GMOs was used both within the Project Accounting Area and Project Zone.

5.1.8 Inputs Justification (B2.8)

Not applicable

5.2 Offsite Biodiversity Impacts

5.2.1 Negative Offsite Biodiversity Impacts (B3.1) and Mitigation Actions (B3.2)

The conservation itself as the aim of the project is already a mitigation strategy. The entire area has greatly benefited since there is no activity involving any kind of human disturbance. Furthermore, conservation of the project area increases landscape integrity and adaptation, avoiding edge effect, as described in the "with project" scenario, benefiting biodiversity (Wunder 2008).

A representative conservation area in which biodiversity can persist guarantees the maintenance of ecological processes and contributes to avoid fragmentation of the ecosystem, both through timber extraction and agricultural activities. The project helps landscapes enhancing its ecological health, including

its adaptability to climate change and consequently reducing offsite greenhouse gas emissions (Wunder 2008). Moreover, the conservation of this area maintains microclimate, avoiding wildfires (Soares-Filho 2006).

Hence as there are no offsite negative biodiversity impacts observed during this monitoring period, there are no planned mitigation measures.

5.2.2 Net Offsite Biodiversity Benefits (B3.3)

The Project is expected to generate positive leakage on biodiversity by avoiding ecosystem fragmentation through voluntary engagement of neighbor communities in the Project's activities. As described above, the project focus exclusively on conservation measures within the project boundaries and its buffer, which makes negative offsite effects unlikely to happen. Besides, monitoring of flora and fauna assures that any minimal offsite negative effect was taken care of immediately. Also, as mentioned on G3.2, the Project's activities do not involve the introduction

Of non-native species. Therefore, considering these activities and "with project" scenario, the effects of the project on biodiversity is positive for this monitoring period.

5.3 Biodiversity Impact Monitoring

5.3.1 Biodiversity Monitoring Plan (B4.1, B4.2, GL1.4, GL3.4)

For presenting biodiversity impact monitoring's results we have created the Table 9 below, that takes into account all biodiversity variables identified in the monitoring plan, dates, frequency, locations, sampling methods used and finally the results and the evaluation of monitoring.

Table 9: key impact factors for the community and other stakeholders.

No.	Variables to be monitored	Results and evaluation	Sampling method and location	Frequency
1	Total planned area for conservation	The planned conservation areas (Project Area) corresponds to the initial planted area of 199,962ha. No losses occurred.	Measuring conserved areas in Project Area.	Each verification period
2	Tree density and diversity	The average number of trees encountered per plot is 40 (each plot being 325.06 m, this means that the average per hectare today is 650 trees/ha.	Permanent samples plots in Project Area.	Each verification period
3	Reduction in soil erosion	There are no more erosions within the Project Area and also within the Project Zone.	Observations in Project Area and Zone	Each verification period
4	Increased natural regeneration in Legal Reserve Zones	During the 5 years of this MP there has been an increase of natural regeneration of the Legal Reserve Zones.	Visual analysis	Each verification period.
5	Increase in biodiversity (FLORA)	- Formation of a "mosaic of ecosystems" in the Project Zone composed due to the conservation of native forest cover - Regeneration of the surrounding Reserve Areas.	Sampling plots in Project Area	Continuous. The registries will be compiled and reported every verification period

6	Increase in biodiversity (FAUNA)	In the Project Zone live permanently populations of 2 species considered "Vulnerable", according to the IUCN Red List: the Jaguar and the Giant Anteater. Also 4 species belong to the "Near threatened" category, are present: the Monkey, the Forest Deer, and the wolf. Other species are present in the Project Zone and their presence from the beginning of the Project to date has had a considerable increase.	Observations in Project Area and Zone.	Continuous. The registries will be compiled and reported every verification period
7	Frequency and intensity of fires	During the 5 years of this Monitoring Period no fire events were recorded within the Project Area, nor in the Legal Reserve Areas.	Observations in Project Area and Zone.	Continuous. Every verification period

5.3.2 Biodiversity Monitoring Plan Dissemination (B4.3)

A summary of the monitoring plan was translated to local language and disseminated to the community groups and other stakeholders prior to validation. Monitoring results was also communicated through meetings and also uploaded on the project website.

5.4 Optional Criterion: Exceptional Biodiversity Benefits

Not applicable.